

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
 - a. Task A: 1 -5
 - b. Task B: 2
2. Domain 2: Therapeutics and Patient Management
 - a. Task A: 1 -6
 - b. Task B: 1, 4, 13, 14, 16
 - c. Task C: 1, 4
3. Domain 3: Professional Practice
 - a. Task A: 4
 - b. Task B: 1, 4, 6

I. PAIN, AGITATION/SEDATION, DELIRIUM, IMMOBILITY, AND SLEEP DISRUPTION (PADIS) IN THE INTENSIVE CARE UNIT

A. Background

1. The Society of Critical Care Medicine (SCCM) published updated guidelines for the management of PADIS in adult ICU patients in 2018 (Crit Care Med 2018;46:e825-e873). These guidelines, together with recently published research, help guide ICU clinicians in the challenging task of optimizing patient comfort and outcomes while avoiding the complications of under- or oversedation. The PADIS guidelines were written by 32 international experts, four methodologists, and four critical illness survivors who met virtually monthly and annually at annual SCCM congresses. Rigorous research has developed our understanding of the assessment tools and medications used for PADIS, the prevention and treatment methods used for PADIS, and the long-term effects of the ICU environment on patients and caregivers. Recommendations for specific ICU populations such as burn, neurologic, neurosurgical (including traumatic brain injury), and cardiac populations may need specialized consideration.
2. The PADIS guidelines add to the 2013 pain, agitation, and delirium (PAD) guidelines by:
 - a. Adding two critical care topics: rehabilitation/mobilization and sleep disruption
 - b. Including patients as collaborators and coauthors
 - c. Inviting an international panel of experts from high-income countries to incorporate more diverse practices and expertise from the global critical care community
3. Each section of the 2018 PADIS guidelines was represented by content experts, methodologists, and ICU survivors. Population, Intervention, Comparison, and Outcome and nonactionable descriptive questions were developed by each section. Members of the guideline group voted on their ranking, and patients voted on their perceived importance. According to the Grading of Recommendations Assessment, Development, and Evaluation principles, every Population, Intervention, Comparison, and Outcomes question was evaluated by members of each section to determine the best level of evidence, assess quality, and determine recommendations as “strong,” “conditional,” or “good” practice. Evidence gaps and clinical caveats were explicitly identified.

- B. Pharmacy Intervention – Pharmacists provide unique and valuable insight into the management of PADIS in the ICU. Much of the management for PADIS involves medications with complex pharmacologic profiles, allowing many opportunities for pharmacy expertise on the critical care team. As the management of PADIS in the ICU continues to evolve, pharmacists should seek avenues for contributing to the critical care community through development of hospital protocols and assessing for quality improvement; providing education for medical, pharmacy, and nursing colleagues; and/or doing research on pertinent questions surrounding the management of PADIS in the ICU (Crit Care Med. 2020;48:e813-e834).

II. PAIN IN THE INTENSIVE CARE UNIT

A. Introduction

1. More than half of ICU survivors report severe pain as the most traumatic memory of their ICU stay.
2. Both short- and long-term negative sequelae are related to uncontrolled pain in the ICU. Assessing pain in the ICU is challenging, particularly in patients who cannot effectively communicate. If patients cannot adequately communicate their degree of pain but retain motor activity, medications should be titrated according to validated behavioral pain scales.

- B. Incidence and Causes of Pain: Pain may occur in any type of ICU patient, and considerations for pain management often require an individualized approach to optimize treatment. The interdisciplinary team should complete a comprehensive review of all variables such as acute and chronic pain, routine nursing care that may cause discomfort, and procedural-based pain.
1. Common causes of pain in the ICU include, but are not limited to, acute trauma, injury or burns, postoperative pain, exacerbation of chronic pain, heart disease, ischemia, acute or chronic underlying disease state pain such as cancer pain, pancreatitis, or other abdominal pathology.
 2. Less discernible causes of pain may include those from either routine nursing care or the provision of life-sustaining measures: presence of an endotracheal tube and endotracheal tube suctioning, wound care, tube or Foley insertion, immobility, bed repositioning, bathing, medication administration, and physical and occupational therapy. Other examples of painful invasive procedures include intravenous line placement, endoscopy and bronchoscopy, chest tube placement or removal, paracentesis, lumbar puncture, biopsies, and fracture reductions.
- C. Short- and Long-term Consequences of Pain in the ICU
1. Acute pain can invoke a stress response, resulting in a hypercatabolic state, decreased tissue perfusion, and impaired wound healing. Uncontrolled pain decreases a patient's immune response to infection by suppressing natural killer cell activity and neutrophil function.
 2. Long-term studies (12 months post-ICU stay) report detrimental physiologic and psychological function in patients who recall significant pain during their hospitalization, particularly in patients admitted with a traumatic injury (Lancet Respir Med 2014;2:369-79).
 - a. Health-related quality of life is decreased in up to 20% of patients.
 - b. Chronic pain is reported in up to 40% of patients.
 - c. Posttraumatic stress disorder is reported in 5%–20% of patients.
- D. Assessment of Pain
1. The gold standard for assessing pain remains the patient's self-report of pain. Several scenarios in the ICU make the self-reporting of pain challenging for clinicians (e.g., mechanical ventilation, presence of sedation and/or delirium). SCCM guidelines currently recommends two validated behavioral pain scales to be done in a repetitive and routine manner: the Behavioral Pain Scale (BPS) (Table 1) and the Critical-Care Pain Observation Tool (CPOT) (Table 2).
 - a. Assessment scales should be used routinely in all ICU patients. Most nursing protocols assess pain every 4–6 hours while the patient is awake. In addition, it is important to reassess the degree of pain within about 30 minutes to 1 hour after administering an "as-needed" pain medication to determine the appropriateness of the pain medication or dose.
 - b. Pain scores should be documented in the medical chart and then used to help formulate daily titrations in pain medications.
 - c. Patients should be treated within 30 minutes of a "significant pain" score. A BPS greater than 5 or a CPOT score of 3 or greater, or a numeric rating scale score of 4 or greater is indicative of pain.
 2. The use of vital signs alone is not recommended for assessing pain in the ICU patient. Abnormal vital signs such as tachycardia and hypertension are appropriate for use as a prompt to further investigate the need for pain control.
 3. Further research is needed to determine the effectiveness of a preprocedural pain assessment tool and the ways in which this assessment will affect analgesic administration. A study by Puntillo et al. in 2014 found the procedures most likely to double the patient's pain intensity score (from preprocedure to during-procedure scoring) were chest tube removal, wound drain removal, and arterial line insertion. This study found that higher-intensity pain and pain distress before the procedure were associated with a high risk of increased pain during the procedure (Am J Respir Crit Care Med 2014;189:39-47).

4. Kanji et al. found that the CPOT is a valid pain assessment in noncomatose, delirious adult ICU patients who are not able to reliably self-report the absence or presence of pain (Crit Care Med 2016;44:943-7).

Table 1. Behavioral Pain Scale (BPS)^a

Item	Description	Score
Facial expression	Relaxed	1
	Partly tightened (e.g., brow lowering)	2
	Fully tightened (e.g., eyelid closing)	3
	Grimacing	4
Upper limbs	No movement	1
	Partly bent	2
	Fully bent with finger flexion	3
	Permanently retracted	4
Compliance with ventilation	Tolerating movement	1
	Coughing but tolerating ventilation most of the time	2
	Fighting ventilator	3
	Unable to control ventilation	4

^aA BPS score > 5 indicates significant pain.

Adapted with permission from: Lippincott Williams & Wilkins/Wolters Kluwer Health. Payen J, Bru O, Bosson J, et al. Assessing pain in critically ill sedated patients by using a behavioral pain scale. Crit Care Med 2001;29:2258-63.

Table 2. Critical-Care Pain Observation Tool (CPOT)^a

Indicator	Description	Score	
Facial expression	No muscular tension observed	Relaxed, neutral	0
	Presence of frowning, brow lowering, orbit tightening, and levator contraction	Tense	1
	All of the above facial movements plus eyelids tightly closed	Grimacing	2
Body movements	Does not move at all (does not necessarily mean absence of pain)	Absence of movement	0
	Slow, cautious movements; touching or rubbing the pain site; seeking attention through movements	Protection	1
	Pulling tube, trying to sit up, moving limbs/ thrashing, not following commands, striking at staff, trying to climb out of bed	Restlessness	2
Muscle tension Evaluation by passive flexion and extension of upper extremities	No resistance to passive movements	Relaxed	0
	Resistance to passive movements	Tense, rigid	1
	Strong resistance to passive movements, inability to complete them	Very tense or rigid	2
Compliance with the ventilator (intubated patients)	Alarms not activated, easy ventilation	Tolerating ventilator or movement	0
	Alarms stop spontaneously	Coughing but tolerating	1
	Asynchrony: Blocking ventilation, alarms often activated	Fighting ventilator	2

Table 2. Critical-Care Pain Observation Tool (CPOT)^a (*continued*)

Indicator	Description	Score	
OR Vocalization (extubated patients)	Talking in normal tone or no sound	Talking in normal tone or no sound	0
	Sighing, moaning	Sighing, moaning	1
	Crying out, sobbing	Crying out, sobbing	2
Total, range			0–8

^aA CPOT score ≥ 3 indicates significant pain.

Adapted with permission from: Lippincott Williams and Wilkins/Wolters Kluwer Health. Gelinas C, Fillion L, Puntillo K, et al. Validation of the Critical-Care Pain Observation Tool in adult patients. *Am J Crit Care* 2006;15:420-7.

E. Prevention and Treatment of Pain in the ICU

1. In a patient whose pain is inadequately controlled in the ICU, intravenous opioids are considered first-line treatment for nonneuropathic pain. The PADIS guidelines suggest acetaminophen as an adjunct to an opioid to decrease pain intensity and opioid consumption for pain management in critically ill patients.
2. Preprocedural pain management should be considered in all ICU patients. One study reported that up to 60% of patients did not receive preprocedural systemic pain medication for common procedures and wound care in the ICU, although 89% of patients received a topical anesthetic for central venous catheter placement (*Am J Crit Care* 2002;11:415-29).
 - a. The PADIS guidelines suggest using an opioid at the lowest effective dose for procedural pain management in critically ill patients undergoing a procedure. The American Society for Pain Management Nursing (ASPMN) published recommendations for preprocedural pain management in 2011. The ASPMN recognizes both the psychological and the physical elements of procedural pain and agrees with combining nonpharmacologic and pharmacologic methods. Examples of nonpharmacologic options recommended by ASPMN include relaxation and breathing techniques, imagery, massage, music, thermal measures, and positioning (*Pain Manag Nurs* 2011;12:95-111). The PADIS guidelines recommend not using inhaled volatile anesthetics for procedural pain management in critically ill adults. The PADIS guidelines also suggest using a nonsteroidal anti-inflammatory drug (NSAID) administered intravenously, orally, or rectally as an alternative to opioids for pain management during discrete and infrequent procedures in critically ill adults. Use of an NSAID topical gel for procedural pain management in critically ill adults is not recommended by the PADIS guidelines.
 - b. Preemptive analgesia for chest tube removal is recommended together with nonpharmacologic relaxation techniques. The PADIS guidelines suggest neither local analgesia nor nitrous oxide for pain management during chest tube removal in critically ill patients.
3. Postoperative thoracic epidural anesthesia/analgesia is recommended for patients undergoing abdominal aortic aneurysm treatment. Thoracic epidural anesthesia is “suggested” for traumatic rib fractures in the ICU.
4. Nonpharmacologic therapy for pain
 - a. The PADIS guidelines suggest offering cold therapy for procedural pain management in critically ill adults.
 - b. The PADIS guidelines suggest offering relaxation techniques for procedural pain management in critically ill adults. Music therapy is suggested for both nonprocedural and procedural pain in critically ill adults.

5. Pharmacotherapy for pain

- a. Intravenous opioids on an as-needed, scheduled, or continuous infusion basis are recommended to treat pain in the ICU. The pharmacokinetics of different opioids may vary; thus, opioids should be chosen according to patient comorbidities and individual needs (Table 3). Fentanyl is the most commonly used intravenous opioids in American adult ICUs.

Table 3. Opiates Commonly Used in the ICU

Drug	Metabolic/Drug Interaction Considerations	Usual CI Starting Dose ^a	Drug-Specific Adverse Effects ^b	Drug Accumulation Factors
Fentanyl	3A4 major substrate	12.5–25 mcg/hr; 0.35–0.5 mcg/kg loading dose followed by 12.5–25 mcg/hr	Muscle rigidity	Hepatic failure; high volume of distribution; high lipophilicity; unpredictable clearance (long context-sensitive half-time) with prolonged infusion
Morphine	Glucuronidation	1–2 mg/hr	Hypotension, bradycardia from histamine release	Hepatic failure; active metabolite (3- and 6-morphine glucuronide) accumulates in renal failure
Hydromorphone	Glucuronidation	0.25–0.5 mg/hr	Overdose effects from dosing errors (high-potency opiate)	Hepatic failure
Methadone	3A4 and 2B6 major substrates	N/A	QTc prolongation, serotonin syndrome	Long half-life; hepatic and renal failure will delay clearance
Remifentanyl	Blood and tissue esterases	Loading dose: 1.5 mcg/kg CI: 0.5–15 mcg/kg/hr	Chest wall rigidity; rebound pain on discontinuation	

^aUsual starting dose in the ICU for pain management in an opiate-naïve patient.

^bOther common significant adverse effects for all opiates to be considered: Constipation, respiratory depression, bradycardia, hypotension, altered mental status.

CI = continuous infusion.

- i. General mechanism of action of opiates: Bind to mu-opioid receptors in the central nervous system (CNS)
- ii. Commonly used intravenous opioids in the ICU: Fentanyl, morphine, hydromorphone, remifentanyl, and methadone
- iii. Tolerance: May quickly develop to all opiates, particularly when given as a continuous infusion. If switching to a different intravenous or oral opiate, equianalgesic dosing may be difficult to estimate, and low starting doses should be considered.
- iv. Significant adverse effects: Decreased respiratory drive: This may be a desired effect in some ICU scenarios; however, a depressed respiratory drive is a critical negative implication during the ventilator weaning process; decreased blood pressure and heart rate, constipation, gastrointestinal (GI) intolerance, altered sensorium.

- v. GI intolerance and constipation: A bowel regimen should be initiated on day 1 unless contraindicated, with assessment for efficacy every 24–48 hours. GI intolerance in the ICU can result in increased time on mechanical ventilation, delayed time to attaining nutritional goals, and prolonged ICU stay. Constipation may also contribute to agitation.
- vi. Altered mental status: Opiates may induce a sedative effect as well as an altered sensorium in some patients. Unless contraindicated, clinicians should consider tapering the opiate dose in an altered patient who has adequate pain control.
- vii. Patient-controlled analgesia: In alert and clinically stable patients, use of patient-controlled analgesia may be considered to titrate to the patient's perceived level of pain. Patient-controlled analgesia may also be useful on discontinuation of continuous infusion opiates.
- b. Fentanyl
 - i. Pharmacokinetics: Hepatic metabolism, cytochrome P450 (CYP) 3A4 substrate. Quick onset and short duration of action; lacks a pharmacologically active metabolite. Highly lipophilic, high volume of distribution and protein binding; maintains a three-compartment model; continuous infusion dosing may lead to prolonged and unpredictable clearance (prolonged context-sensitive half-time).
 - ii. Many dosage forms: Injectable (intravenous, intramuscular, intrathecal, epidural), transdermal, transmucosal, nasal spray. Different dosage forms should not be converted on a 1:1 mcg basis; use specific manufacturer recommendations if converting. Injectable form of fentanyl is most commonly used in the ICU setting. The fentanyl patch is not generally appropriate for use in the ICU because of its latent onset (about 12 hours) and erratic/increased absorption in a febrile patient.
 - iii. Adverse effects: Respiratory depression, bradycardia, hypotension, CNS depression, constipation, ileus, risk of serotonin syndrome when used with other serotonergic agents
- c. Morphine
 - i. Pharmacokinetics: Hepatic metabolism by glucuronidation to two major active metabolites, morphine-3-glucuronide (45%–55%) and morphine-6-glucuronide (10%–15%). The glucuronide metabolites of morphine are both renally eliminated; accumulation can occur with the chronic use of morphine or in patients with decreased renal function. Morphine-3-glucuronide does not have analgesic activity, but adverse effects may include seizure activity or agitation. Morphine-6-glucuronide does have analgesic activity by the mu-receptor and may cause additive sedation and respiratory depression if accumulation occurs. Continuous morphine infusions are rarely used for analgesia in the ICU setting because of the concerns with the active metabolites.
 - ii. Dosage forms: Injectable (intravenous, subcutaneous, intrathecal, epidural) and oral. Intravenous-to-oral conversion is not a 1:1 mg ratio.
 - iii. Adverse effects: Histamine release may cause significant hypotension; bradycardia, respiratory depression, CNS depression, constipation, ileus.
- d. Hydromorphone
 - i. Pharmacokinetics: Hepatic metabolism by glucuronidation to an inactive but potentially neurotoxic metabolite. Low volume of distribution, highly water soluble, and relatively low protein binding.
 - ii. Dosage forms: Injectable (intravenous, subcutaneous) and oral; intravenous-to-oral conversion is not a 1:1 mg ratio.
 - iii. Adverse effects: CNS alterations (e.g., abnormal dreams, aggressive behavior, altered thinking), respiratory depression, hypotension, constipation

- e. Remifentanyl (Ultiva)
 - i. Pharmacokinetics: Clearance by blood and tissue esterase; clearance not dependent on organ function. Fast onset and short duration of action with little to no accumulation. High volume of distribution, high protein binding.
 - ii. Research primarily done in Europe; limited reported use in U.S. adult ICUs for ongoing analgesic use.
 - iii. Dosage form: Injectable only
 - iv. Adverse effects: Respiratory depression, hypotension, bradycardia, constipation
 - v. Rebound pain: Quick offset (5–10 minutes) may lead to rebound pain and withdrawal symptoms, and additional pain medication may be needed if remifentanyl is interrupted or discontinued.
 - vi. Benefit in adult ICUs: Decreased time on mechanical ventilation with short-term use (72 hours or less)
- f. Methadone
 - i. Pharmacokinetics: Phase I hepatic metabolism to inactive metabolites. Many drug interactions: major substrate of CYP 2B6 and 3A4. Moderate inhibitor of CYP2D6, weak inhibitor of CYP3A4. Longer-acting opiate with variable duration of action (12–48 hours); may accumulate quickly in patients with hepatic failure or patients receiving hemodialysis. Animal studies have found that the d-isomer of methadone works as both a partial mu-agonist and an N-methyl-d-aspartate receptor antagonist (the l-isomer is a full mu-agonist). These properties of the d-isomer are thought to decrease the tolerance effect to other opioids. Methadone is currently marketed as the racemic mixture. On initiating oral methadone, steady state and peak analgesic effect may not be reached for 3–5 days; oversedation and respiratory depression may occur if titrated too quickly.
 - ii. Dosage forms: Injectable (intravenous, intramuscular, subcutaneous) and oral. Intravenous-to-oral conversion is not a 1:1.3 mg ratio.
 - iii. Adverse effects: Dose-dependent QTc prolongation, altered mental status, respiratory depression, confusion, dizziness, arrhythmias, constipation, risk of serotonin syndrome when used with other serotonergic agents
- 6. Non-opioid adjunctive pain medications should be considered in combination with opioids to reduce opioid requirements. Clinically stable patients may tolerate a conversion from opiates to non-opiate medications.
 - a. Local and regional anesthetics such as bupivacaine. The PADIS guidelines do not recommend the use of intravenous lidocaine as an adjunct to opioid therapy for pain management in critically ill adults.
 - b. Acetaminophen (Tylenol, Ofirmev): The PADIS guidelines suggest using acetaminophen as an adjunct to opioids to decrease pain intensity and opioid consumption.
 - i. Total daily acetaminophen doses should be considered from all acetaminophen combination products, with a maximum total daily dose of 4 g. Decreased total daily dosing should be considered in patients with significant liver disease or those older than 60 years of age.
 - ii. Intravenous acetaminophen: Dose reduction recommended if the creatinine clearance (CrCl) is 30 mL/minute/1.73 m² or less or with continuous renal replacement therapy (every 8 hours); contraindicated in severe hepatic disease. The cost of the intravenous formulation of acetaminophen is considerably higher than that of the oral or rectal formulations. The intravenous formulation can also cause hypotension.
 - c. Intravenous or oral NSAIDs: Ibuprofen, ketorolac. The PADIS guidelines suggest not routinely using a COX-1 selective NSAID as an adjunct to opioid therapy for pain management in critically ill adults. Use with caution in critically ill patients with renal or hepatic dysfunction. May increase the risk of acute renal failure, bleeding, or GI adverse effects.

- d. Ketamine (Ketalar) has been used for analgesia and sedation in the ICU, primarily in the pediatric population. The PADIS guidelines suggest using low-dose ketamine (1–2 mcg/kg/minute) as an adjunct to opioid therapy when seeking to reduce opioid consumption in postsurgical adults admitted to the ICU. Published data for the use of ketamine in adults for analgesia and/or sedation are limited to case reviews, and long-term cognitive effects of ketamine are not known. Data from animal studies suggest a significant decline in cognitive function after continued use of ketamine.
 - i. Called a “dissociative anesthetic,” providing analgesic activity at subanesthetic doses. It is a schedule III controlled substance and works primarily as an N-methyl-D-aspartate receptor antagonist. Ketamine is void of the constipation, respiratory depression, and hypotensive effects that plague the opiate class.
 - ii. May decrease dose requirements of concurrently administered opioids
 - iii. Other uses include rapid sequence intubation, refractory pain syndromes, cancer pain, neuropathic pain, asthma (bronchodilatory effects), refractory seizure activity, and depression.
 - iv. Dosing range is varied; usual starting dose for analgesia or sedation is 0.1 mg/kg/hour. Reviews of ketamine use in adult ICUs report a dosing range of 0.1–2.5 mg/kg/hour and a range in duration of 3 hours to 9 days.
 - v. Significant adverse effects: Mild to severe emergence reactions (e.g., confusion, excitement, irrational behavior, hallucinations, delirium) in around 12% of patients, enhanced skeletal muscle tone, tachycardia, hypertension, hypotension
 - e. The PADIS guidelines suggest not offering cybertherapy or hypnosis for pain management in critically ill adults.
 - f. The PADIS guidelines suggest offering massage for pain management in critically ill adults.
 - g. The PADIS guidelines suggest offering music therapy to relieve both non-procedural and procedural pain in critically ill adults.
7. The PADIS guidelines suggest using a neuropathic pain medication (e.g., gabapentin, carbamazepine, pregabalin) with opioids for neuropathic pain management in critically ill patients. The PADIS guidelines recommend using neuropathic pain medications with opioids for pain management in ICU adults after cardiovascular surgery. There is a potential for significant adverse effects and drug interactions, requiring close monitoring and follow-up. If the patient is discharged home on an anticonvulsant for neuropathic pain, follow-up should be documented and the primary care provider notified.
- a. Gabapentin (Neurontin)
 - i. Suggested starting dose range: 300–600 mg/day divided two or three times daily; requires renal adjustment. The target dose is 900–3600 mg/day in three divided doses.
 - ii. Pharmacokinetics: Renally excreted, dose adjusted for reduced CrCl
 - iii. Adverse effects: May be severe, including CNS depression, delirium, paresthesias, and asthenias
 - b. Carbamazepine (Tegretol)
 - i. Suggested starting dose range: 50–100 mg twice daily; use with caution in patients with hepatic impairment, and adjust for a CrCl less than 10 mL/minute/1.73 m² or with hemodialysis. The target dose is 100–200 mg every 4–6 hours; the maximum dose is 1200 mg/day.
 - ii. Pharmacokinetics: Strong inducer of many CYP enzymes, substrate of CYP3A4. Closely monitor for drug interactions.
 - iii. Adverse effects: Somnolence, severe skin reactions (e.g., Stevens-Johnson syndrome, toxic epidermal necrolysis), pancytopenia, syndrome of inappropriate antidiuretic hormone
 - c. Pregabalin (Lyrica)
 - i. 50 mg three times daily; may be increased in 1 week depending on tolerability and effect; maximum dose: 100 mg three times daily.
 - ii. Pharmacokinetics: 90% excretion in urine

- iii. Adverse effects: Peripheral edema, dizziness, drowsiness, headache, fatigue, weight gain, xerostomia, visual field loss, and blurred vision
- F. Analgesedation Method in the ICU: This method of sedation advocates the use of opiate medications before prescribing an anxiolytic/hypnotic medication to provide patient comfort in the ICU unless anxiolytics are otherwise indicated. The PADIS guidelines recommend analgesedation (analgesia is used before a sedative to reach the sedative goal) or analgesia-based sedation (an opioid is used instead of a sedative to reach the sedative goal). Providing pain relief early in the ICU stay may decrease the agitation associated with pain and/or general discomfort while minimizing the use of alternative medications commonly used for agitation (e.g., benzodiazepines). The guidelines recognize that current data using analgesedation are primarily limited to open-label trials, using remifentanyl as the analgesic, and mostly conducted in Europe, where critical care staffing and management practices differ from those in the United States. Despite these limitations, it remains notable that studies using the analgesedation method found a significant decrease in benzodiazepine dosage requirements when opiates were the primary medications used for discomfort and agitation. This is a positive step in decreasing the untoward adverse effects of the benzodiazepine class of sedatives. There is a potential for high cumulative doses of opiates with the analgesedation method, necessitating daily monitoring of their adverse effects (e.g., respiratory depression, altered mental status, GI slowing). Opioid use in the ICU also increases the daily risk of delirium development in a dose dependent manner.
- G. Data: NONSEDA is a randomized, clinical parallel-group, multinational superiority study comparing no sedation (bolus doses of morphine) and light sedation (propofol for 48 hours then midazolam titrated to a RASS of -2 to -3). There was no difference in 90-day mortality between groups (no sedation 148 [42.4%] vs. light sedation 130 [37%]; 95% CI -2.2 to 12.2; $p=0.65$). However, 27% of the no-sedation group crossed over to receive sedation during the first 24 hours after randomization.

Patient Case

Questions 1 and 2 pertain to the following case.

T.O. is a 70-year-old man just admitted to the ICU with multiple fractures after a motor vehicle accident. His medical history includes hypertension. He is now agitated after intubation. His laboratory values are normal, and his vital signs include blood pressure 175/95 mm Hg and heart rate 110 beats/minute.

1. Which grouping of initial sedatives is most appropriate at this time?
 - A. Fentanyl infusion and midazolam infusion
 - B. Propofol infusion and fentanyl as needed
 - C. Midazolam as needed and fentanyl as needed
 - D. Fentanyl infusion and propofol infusion, if needed
2. After 2 weeks in the ICU, T.O. is being prepared for chest tube removal. He is currently receiving a fentanyl drip with adequate pain control. Which is the best pain management regimen for chest tube removal?
 - A. Give intravenous acetaminophen 15 minutes before chest tube removal.
 - B. Make no change in pain treatment because his current pain regimen is adequate.
 - C. Increase his pain medication infusion dose by 50% the morning of his chest tube removal.
 - D. Give fentanyl 50 mcg injectable 15 minutes before chest tube removal.

III. AGITATION IN THE INTENSIVE CARE UNIT

- A. Agitation in the ICU – Maintaining patient comfort for the duration of an ICU stay can be extremely challenging, requiring significant resources and daily discipline from the nursing, medical, and pharmacy team. Ongoing research has improved our understanding of the consequences of either under- or overtreating agitation in the ICU, and clinicians should continue to apply this knowledge to their daily selection and titration of medications. Treatment of a patient who presents with agitation must always begin with attempts to identify and correct the etiology of the agitation. Common causes of agitation in the ICU include pain, delirium, hypoxia, hypoglycemia, dehydration, and drug or alcohol withdrawal. Close inspection of significant patient variables will also help determine the appropriate sedative:
1. Pain control
 2. Substance abuse and smoking history
 3. Neurologic function: Baseline and acute mental status, history of seizure activity, dementia, psychiatric history
 4. Clinical variables: Blood pressure, heart rate, respiratory rate
 5. Comorbidities (baseline and acute): Cardiac, renal, hepatic, gastric, pulmonary, pancreatic
 6. Home medication use: Any medication from which a patient could withdraw: Benzodiazepines, opioids, nicotine, antidepressants, other γ -aminobutyric acid (GABA) receptor agonists
- B. Common Medications for the Treatment of Agitation – Include propofol, dexmedetomidine, and benzodiazepines (usually lorazepam and midazolam) (Table 4). Benzodiazepines are first-line agents for status epilepticus, alcohol withdrawal, benzodiazepine dependence or withdrawal, and are useful for deep sedation or amnesia and with the use of neuromuscular blockade. Other indications for benzodiazepines may exist, which must be scrutinized throughout the ICU stay.

Table 4. Sedatives for Patients on Mechanical Ventilation in the ICU

Drug	Onset and Duration	Precautions for Use	CYP Substrate (major)	Usual Dose	Significant Adverse Effects
Propofol	Onset: 1 min Duration: short term: 0.5–1 hr; long term > 7 days: variable; 25–50 hr has been observed (depends on depth and time on sedation)	Hypotension, bradycardia, hepatic/renal failure, pancreatitis	2B6	5–50 mcg/kg/min; 0.3–3 mg/kg/hr	Hypotension, pancreatitis, respiratory depression, bradycardia, PRIS
Dexmedetomidine	Onset: 5–10 min (with LD) 1–2 hr (without LD) Duration: 1–2 hr	Hepatic failure; symptomatic bradycardia	2A6	LD: 0.5–1 mcg/kg (optional) MD: 0.2–0.7 mcg/kg/hr	Hypo/hypertension, bradycardia, pyrexia
Lorazepam	Onset: 5–20 min Duration: 4–8 hr; prolonged with continuous infusion	Delirium, renal failure	N/A	Intermittent: 1–4 mg IV every 4–6 hr	Oversedation, propylene glycol toxicity
Midazolam	Onset: 3–5 min Duration: 2–6 hr, prolonged with continuous infusion	Hepatic failure, end-stage renal failure or dialysis, delirium	3A4 (active metabolite)	0.02–0.1 mg/kg/hr	Oversedation

CI = continuous infusion; IV = intravenously; LD = loading dose; MD = maintenance dose; N/A = not applicable; PRIS = propofol-related infusion syndrome.

- C. SCCM provides the following statement in the PADIS guidelines regarding sedation in the ICU: “We suggest using either propofol or dexmedetomidine over benzodiazepines [midazolam or lorazepam] for sedation in critically ill, mechanically ventilated adults” and those after cardiac surgery. SCCM states that “benzodiazepine use may be a risk factor for the development of delirium in adult ICU patients.” Two randomized studies evaluated the differences in clinical outcomes while adult ICU patients were receiving sedation with either a benzodiazepine or a non-benzodiazepine strategy. Heterogeneity occurred among the findings of these two studies (see No. 1 and No. 2 below), which may be partly because of differences in study design.
1. The MENDS study compared the sedative effects of lorazepam and dexmedetomidine in medical and surgical adult ICU patients (n=103). Dexmedetomidine had more “delirium-free + coma-free” days than lorazepam (7 vs. 3 days, $p=0.01$), and the prevalence of “delirium or coma” was lower in the dexmedetomidine group (87% vs. 98%, $p=0.03$). The assessment of “delirium-free + coma-free” was a more appropriate outcome to evaluate than “delirium without coma,” given that delirium cannot be assessed in patients with coma. More patients were within 1 point of their RASS goal with dexmedetomidine (67%) than with lorazepam (55%), $p=0.008$, but there was no difference in mechanical ventilator-free days, ICU length of stay, or 28-day mortality. This study has been criticized because both groups received continuous infusions of sedatives without additional bolusing, whereas in clinical practice, most practitioners would bolus lorazepam before increasing the infusion rate. In addition, patients were not required to have a spontaneous awakening trial (SAT). (JAMA 2007;298:2644-53).
 2. The SEDCOM study compared the sedative effects of midazolam with those of dexmedetomidine in medical and surgical adult ICU patients (n=366). The prevalence of delirium was lower in the dexmedetomidine group (54%) than in the midazolam group (76.6%), $p<0.001$. Median time to extubation was shorter in the dexmedetomidine group (3.7 days) than in the midazolam group (5.6 days), $p<0.01$; however, the times in target sedation range, ICU length of stay, and mortality were no different between the two groups. This study did allow bolus dosing of study drugs, and patients were required to have an SAT if safety criteria were met (JAMA 2009;301:489-99).
- D. Clinical Outcome Differences Among Sedative Agents in Medical and Surgical ICU Patients: A meta-analysis, “Benzodiazepine versus Nonbenzodiazepine-Based Sedation for Mechanically Ventilated Critically Ill Adults,” reviewed trials from 1996 to 2013 (Crit Care Med 2013;41:S30-8):
1. Studies from this review contained the following criteria: (1) randomized controlled parallel-group design; (2) medical and surgical adult ICU patients on mechanical ventilation receiving intravenous sedation; (3) patients receiving a non-benzodiazepine (propofol 1% or dexmedetomidine) compared with a benzodiazepine (lorazepam or midazolam); and (4) patients having predefined outcomes. Excluded cardiac surgery and obstetric patients.
 2. Four primary outcomes from six randomized trials were reported in the review (1235 patients):
 - a. ICU length of stay (all six studies reported): ICU length of stay was longer in a benzodiazepine strategy than in a non-benzodiazepine-based strategy (mean difference 1.6 days; 95% confidence interval [CI], 0.72–2.5; $p=0.0005$).
 - b. Duration of mechanical ventilation (four studies reported): Longer duration of mechanical ventilation in a benzodiazepine-based strategy than in a non-benzodiazepine-based strategy (mean difference 1.9 days; 95% CI, 1.7–2.09; $p=0.00001$)
 - c. Delirium prevalence (two studies reported): No difference in delirium prevalence between a benzodiazepine and a non-benzodiazepine-based strategy (relative risk [RR] 0.98; 95% CI, 0.76–1.27; $p=0.94$)
 - d. Short-term (45 days or less) all-cause mortality (four studies reported): No difference in risk of death between a benzodiazepine and a non-benzodiazepine-based strategy (RR 0.98; 95% CI, 0.76–1.27, $p=0.94$)

E. Propofol

1. SCCM suggests using a non-benzodiazepine (propofol or dexmedetomidine) for sedation to improve clinical outcomes in mechanically ventilated patients. Throughout the past decade, propofol has been increasingly used worldwide for sedation in the ICU. Propofol's short duration of action, lack of accumulation, and relatively clean adverse effect profile at low to moderate doses makes it an appealing alternative.
2. Mechanism of action: General anesthetic by potentiation of the GABA_A receptor; may inhibit *N*-methyl-D-aspartate receptor activity at high doses. Propofol decreases cardiac β -adrenergic responsiveness and attenuates β -adrenergic signal transduction in cardiac myocytes, resulting in direct cardiac depressive effects.
3. Pharmacokinetics: Hepatic conjugation; clearance may be prolonged (from minutes to hours) in patients with severe hepatic impairment or cirrhosis or with long-term infusions as it redistributes from fat and muscle to plasma. Highly lipophilic pharmacokinetics and a large volume of distribution lead to extensive tissue distribution. Propofol maintains a three-compartment linear model: plasma, rapidly equilibrating tissues (e.g., major organs), and slowly equilibrating tissues (e.g., fat deposits). Substrate of CYP 2B6, 2C9, 2C19, and 3A4; pharmacokinetic studies of healthy volunteers show a 25% increase in propofol plasma concentrations when given with midazolam, a weak CYP 3A4 and 2C9 inhibitor.
4. Lipid formulation considerations: Standard propofol is a 1% (10 mg/mL) lipid emulsion containing 1.1 kcal/mL (0.1 g of fat per 1 mL of propofol); this should be accounted for when calculating nutritional intake (e.g., propofol at 50 mcg/kg/minute in a 70-kg patient would provide around 500 calories per day contributed by fat). Propofol contains 0.005% disodium edetate (EDTA) to decrease the rate of microorganism growth, which is known to chelate trace metals, including zinc. Zinc supplementation should be considered in patients at high risk of zinc deficiency (sepsis, burns, large-volume diarrhea) if propofol is used for more than 5 days. Strict aseptic technique must be followed when handling propofol; manufacturers recommend discarding propofol bottles and changing intravenous tubing every 12 hours to decrease the risk of contamination. Propofol is not contraindicated in patients with egg or soybean allergies as historically thought. Propofol contains egg yolks, and most patients are allergic to egg whites. Propofol contains soybean oil not soybean-derived proteins, which have been linked to allergic reactions.
5. Dosing range for ICU sedation: Usual starting dose 5–10 mcg/kg/minute, titrated every 5–10 minutes to goal sedative effect. Abrupt discontinuation of propofol is not recommended because of its rapid clearance (5–10 minutes).
6. Data: In a multicenter European trial (PRODEX), Jakob et al. compared propofol (n=249) with dexmedetomidine (n=251) for sedation in prolonged mechanical ventilation. Patients in both groups were treated with daily sedation interruption trials and spontaneous breathing trials (SBTs), and pain was treated with fentanyl boluses. Proportion of time in target RASS (0, –3) without rescue therapy was the same in the propofol group (65%) as in the dexmedetomidine group (65%). There was no difference in median time on mechanical ventilation in propofol (5 days) versus dexmedetomidine (4 days), $p=0.24$. Patients' ability to communicate discomfort was better in the dexmedetomidine group. Rates of hypotension and bradycardia were similar between the two groups. Critical illness polyneuropathy was more common in the propofol group (n=11) than in the dexmedetomidine group (n=2), $p<0.02$. The composite outcome of agitation, anxiety, and delirium occurred in propofol (29%) versus dexmedetomidine (18%) ($p=0.008$) (JAMA 2012;307:1151-60).
7. Adverse effects: Bradycardia and hypotension (may be more common or severe in patients with cardiac dysfunction, intravascular volume depletion, or low systemic vascular resistance); respiratory depression, hypertriglyceridemia (most institutions use a threshold of 800–1000 as a cutoff of when to stop propofol), pancreatitis with or without hypertriglyceridemia, PRIS

8. PRIS: This is a rare but life-threatening complication of propofol, usually occurring at doses greater than 50 mcg/kg/minute for 48 hours or more. The mechanism of PRIS may include alterations in hepatic metabolism of the lipid emulsion, leading to an accumulation of ketone bodies and lactate and/or disruptions in the mitochondrial respiratory chain and inhibition of oxidative phosphorylation. Patients with urea cycle disorders may experience alterations in propofol metabolism within 24–48 hours of propofol use. Consider avoiding in patients with acute liver failure, or pancreatitis, because the symptoms of PRIS may be difficult to distinguish from the underlying disease state abnormalities. PRIS carries a high mortality rate, and propofol should be discontinued immediately if symptoms are present.
 - a. Clinical characteristics of PRIS: Metabolic acidosis, acute renal failure, cardiovascular collapse, cardiac arrhythmias including Brugada-like syndrome, rhabdomyolysis, myoglobinemia, myoglobinuria, hyperkalemia, hypertriglyceridemia, elevated creatine kinase concentrations
 - b. Risk factors for PRIS or other adverse effects of propofol: Neurologic injury, sepsis, use of vasoactive medications, high-dose propofol, acute liver failure

F. Dexmedetomidine

1. SCCM suggests that in adult ICU patients with delirium unrelated to alcohol or benzodiazepine withdrawal, dexmedetomidine infusions rather than benzodiazepine infusions should be administered for sedation to reduce the duration of delirium.
2. Mechanism of action: Highly selective and dose-dependent α_2 -adrenoceptor agonist in the CNS. Dexmedetomidine provides a hypnotic and sedative effect by inhibition of norepinephrine release from the locus coeruleus; dexmedetomidine also produces a weak antinociceptive effect by way of inhibition of neuronal transmission through presynaptic C-fibers and release of substance P, and hyperpolarization of postsynaptic α receptors in the dorsal horn of the spinal column. Dexmedetomidine does not directly affect respiratory drive; therefore, intubation is not required with use. Dexmedetomidine is considered a weak sedative and would not be appropriate for use when deep sedation is required (e.g., in a patient requiring neuromuscular blockade). Both anterograde amnesia and retrograde amnesia have been described in some patients (20%–50%) in adult and pediatric studies. A benzodiazepine may be required if full amnesia is desired because of the clinical scenario.
3. Pharmacokinetics: Hepatic by glucuronidation and renal excretion. Onset with loading dose 15–20 minutes; onset without loading dose greater than 20 minutes to 1 hour; terminal half-life = 3 hours (may be significantly prolonged in hepatic impairment). Highly protein bound 94%.
4. Clinical effects: Sedation and weak opiate-sparing antinociceptive effects.
5. Dosing for ICU sedation: Optional loading dose 0.5–1 mcg/kg intravenously for 10 minutes, followed by 0.2–0.7 mcg/kg/hour. The loading dose may initially cause severe tachycardia and hypertension, but it can then quickly lead to significant bradycardia and/or hypotension secondary to receptor saturation. Because of these untoward hemodynamic effects, the loading dose is rarely administered in clinical ICU practice on initiation of dexmedetomidine, and the drip is usually initiated at 0.2–0.4 mcg/kg/hour. For the maintenance infusion dose, randomized trials have safely used dexmedetomidine at higher than manufacturer-recommended doses, up to 1.5 mcg/kg/hour. Clinical efficacy with doses greater than 1.5 mcg/kg/hour remains unclear. Other routes of administration have been described for dexmedetomidine, including intramuscular, subcutaneous, epidural, and intranasal.
6. Duration of use: Although the package insert recommends a therapy of 24 hours or less, randomized trials have used dexmedetomidine for up to 5–7 days; thus, ICU clinicians often administer dexmedetomidine for longer than 24 hours. Safety beyond 7 days of use has not been well established.
7. Data: The Dexmedetomidine to Lessen ICU Agitation (DahLIA) study was a double-blind placebo-controlled, parallel-group randomized clinical trial in 15 ICUs in Australia and New Zealand in which 39 patients were randomized to dexmedetomidine and 32 patients to placebo. At 7 days, dexmedetomidine increased ventilator-free hours compared with placebo (median, 144.8 hours vs. 127.5 hours, 95% CI

- 4 to 33.2 hours, $p=0.01$). Patients in the dexmedetomidine group had decreased time to extubation compared with placebo (median 21.9 hours vs. 44.3 hours, 95% CI 5.3 to 3.1 hours, $p<0.001$). An accelerated resolution of delirium was found in the dexmedetomidine group compared with placebo (median, 23 hours vs. 40 hours, 95% CI, 3 to 28 hours, $p=0.01$). However, propofol use was common in both groups after randomization (72% in the dexmedetomidine group vs. 88% in the placebo group) (median cumulative dose 980 mg (IQR 280–3050) vs. 5390 mg (IQR 1880–10,803), $p<0.001$).
8. Data: The DESIRE trial was an open-label, multicenter randomized clinical trial conducted in eight ICUs in Japan in which 100 patients with sepsis were randomized to dexmedetomidine and 101 patients with sepsis were randomized to placebo. Mortality at 28 days was not different between the two groups (23% in the dexmedetomidine group vs. 31% in the placebo group; HR 0.69; 95% CI, 0.38–1.22; $p=0.20$). Ventilator-free days over 28 days were not different between groups (median 20 days (IQR 5–24) in the dexmedetomidine group vs. 18 days (IQR 0.5–23) in the control group ($p=0.20$).
 9. Data: SPICE III was an open-label, randomized controlled trial comparing dexmedetomidine as primary sedation with usual care (propofol, midazolam, or other sedation) in patients receiving less than 12 hours of mechanical ventilation who are expected to be on the ventilator for at least one additional day. The target RASS goal was -2 to +1. Death at 90 days occurred in 569 of 1956 (29.1%) of the usual care group and 566 of 1948 (29.1%) of the dexmedetomidine group (adjusted risk difference 0.0 percentage points; 95% CI, -2.9 to 2.8). In the dexmedetomidine group, 64% of patients additionally received propofol, 3% received midazolam, and 7% received both during the first 2 days after randomization. Sixty percent of patients received propofol, 12% received midazolam, and 20% received both in the usual care group. Given the many sedatives administered in both groups, application of the study results is difficult. In the dexmedetomidine group, bradycardia and hypotension occurred in 5.1% and 2.7% of patients. A post hoc analysis of SPICE III included 703 mechanically ventilated patients. Over the first 5 ICU days, more patients in the dexmedetomidine group had a temperature of 38.3°C or greater (43.3% vs 32.7%, $p=0.004$) and 39.0°C or greater (19.4% vs 12.5%, $p=0.013$). A post hoc subgroup analysis of SPICE III compared the effect of dexmedetomidine versus usual care on vasopressor requirements in patients with septic shock. This subgroup included 83 patients, of which, 44 (53%) received dexmedetomidine and 39 (47%) usual care. In the first 48 hours, median NEq dose was 0.03 [0.01, 0.07] $\mu\text{g/kg/min}$ in the dexmedetomidine group and 0.04 [0.01, 0.16] $\mu\text{g/kg/min}$ in the usual care group ($p = 0.17$). Patients in the dexmedetomidine group had a lower NEq/MAP ratio, meaning lower vasopressor requirements to maintain the target MAP.
 10. Data: MENDS 2 is multicenter, double-blind trial comparing dexmedetomidine to propofol. No difference was found between dexmedetomidine and propofol in the number of days alive and without delirium of coma (adjusted median 10.7 vs 10.8 days; odds ratio, 0.96; 95% CI 0.74–1.26), ventilator free days (adjusted median, 23.07 vs. 24 days, odds ratio 0.98; 95% CI, 0.63–1.51), or death at 90 days 38% vs. 39%; hazard ratio, 1.06; 95% CI 0.74–1.52).
 11. Data: Dexmedetomidine was compared with other sedatives in critically ill mechanically ventilated adults. A total of 77 randomized trials were included. Compared with other sedatives, dexmedetomidine reduced the risk of delirium (RR 0.67; 95% CI, 0.55–0.81; moderate certainty), duration of mechanical ventilation (mean difference [MD] -1.8 hours; 95% CI, -2.89 to -0.71; low certainty), and ICU length of stay (MD -0.32 days; 95% CI, -0.42 to -0.22; low certainty). The risk of bradycardia (RR 2.39; 95% CI, 1.82–3.13; moderate certainty) and hypotension (RR 1.32; 95% CI, 1.07–1.63; low certainty) was increased with dexmedetomidine.
 12. Data: A rapid practice guideline was published looking at use of dexmedetomidine for sedation in mechanically ventilated adult patients in the ICU. It suggested use of dexmedetomidine over other sedative agents if the reduction in delirium was more highly valued than the potential for hypotension and bradycardia.
 13. Adverse effects: Tachycardia, pyrexia bradycardia, hypertension, hypotension, dry mouth. Should generally be avoided in patients with acute decompensated heart failure or advanced heart block

14. Other potential uses in the ICU: Procedural sedation, palliative care pain and anxiety control, adjunct to opiates for sickle cell crisis, adjunct to benzodiazepines or propofol for alcohol withdrawal, bridge to extubation while tapering off longer-acting sedatives and/or opiates, to provide sedation and anxiolysis during noninvasive mechanical ventilation

G. Lorazepam (Ativan)

1. Pharmacokinetics: Benzodiazepine that binds to the postsynaptic GABA_A receptor, undergoes hepatic clearance by conjugation to inactive compounds; moderate to high volume of distribution and high protein binding. Onset of action is 15–30 minutes, slower than more lipophilic benzodiazepines (e.g., midazolam). Duration of action of intermittent dosing is 4–8 hours. As a continuous infusion, clearance of lorazepam decreases in an unpredictable fashion, and prolonged sedation may occur.
2. Effects: Anxiolysis/sedation, anticonvulsant, muscle relaxant. Maintains anterograde amnesia properties; however, studies report that patients who received benzodiazepines in the ICU may maintain delusional versus factual memories.
3. Dosing range: 1–4 mg every 4–6 hours intermittent dosing is recommended before using continuous infusion; accumulation and prolonged awakening times with continuous infusion of lorazepam may occur because of prolonged duration of action.
4. Data: Carson et al. studied the number of days on mechanical ventilation in medical ICU patients receiving intermittent lorazepam (n=64) compared with continuous infusion propofol (n=68); each group underwent daily interruption of sedation if the fraction of inspired oxygen (Fio₂) was less than 80%. Median time on mechanical ventilation was 9 days in the lorazepam group versus 4.4 days in the propofol group, p=0.006; ICU length of stay was 12.7 days in the lorazepam group versus 8.6 days in the propofol group (p=0.05); no difference in hospital mortality. Delirium was not assessed in this study (Crit Care Med 2006;34:1326-32).
5. Propylene glycol toxicity: Because of its insolubility, injectable lorazepam is diluted in propylene glycol. Propylene glycol toxicity can occur with lorazepam infusions for more than 48 hours, particularly at doses of 6–8 mg/hour or greater, and can manifest as new-onset renal failure, respiratory failure, metabolic acidosis, and altered mental status. Most hospitals cannot measure quantitative levels of propylene glycol; therefore, surrogate markers of propylene glycol toxicity such as an elevated osmolar gap (greater than 10 mOsm/kg) and elevated anion gap with new metabolic acidosis are recommended for monitoring. If these metabolic abnormalities are present while on a lorazepam infusion, lorazepam should be discontinued.
6. Other adverse effects: Paradoxical agitation, confusion, prolonged duration of sedative action, respiratory depression, hypotension, bradycardia

H. Midazolam (Versed)

1. Pharmacokinetics: Benzodiazepine that binds to the postsynaptic GABA_A receptor; undergoes phase I hepatic metabolism to an active glucuronidated metabolite, α_1 -hydroxymidazolam, which is then renally excreted. A short- to medium-acting benzodiazepine in patients with normal renal and hepatic function. CYP3A4 substrate. Midazolam is highly lipophilic, has a large volume of distribution, and is highly protein bound.
2. Clearance: Clearance of midazolam or its metabolite is significantly altered if either hepatic (primary drug accumulation) or renal (active metabolite α -hydroxymidazolam accumulation) functions are significantly impaired. Continuous renal replacement therapy partly clears the active metabolite but does not effectively clear the parent compound and is therefore not recommended as a method for definitive midazolam clearance (Am J Kidney Dis 2005;45:360-71). High lipophilicity and a large volume of distribution may lead to significant drug accumulation and a depot effect in the ICU patient. In general, clearances of midazolam infusions have wide interpatient variability in the ICU, and emergence times may be significantly prolonged.

3. Effects: Anxiolysis/sedation, anticonvulsant, muscle relaxant. Maintains anterograde amnesia properties.
4. Dosing range: 1–4 mg every 2–4 hours intermittently or as needed should be considered before initiating continuous infusion. Older adult patients may tolerate only 1–2 mg per dose.
5. Data: In a multicenter European trial (MIDEX), Jakob et al. compared midazolam (n=251) with dexmedetomidine (n=249) for sedation in prolonged mechanical ventilation. Patients in both groups were treated with daily sedation interruption trials and SBTs, and pain was treated with fentanyl boluses. There was no difference in the primary outcome: proportion of time in target RASS (0, -3) without rescue therapy in the midazolam group (56%) versus the dexmedetomidine group (60%). Median time on mechanical ventilation was lower in the dexmedetomidine group (5 days) than in the midazolam group (6.8 days), $p=0.03$. Patients in the dexmedetomidine group were more arousable, more cooperative, and better able to communicate discomfort or pain to clinical staff than were patients in the midazolam group. Hypotension occurred more often in the dexmedetomidine group (20.6%) than in the midazolam group (11.6%; $p=0.007$); and bradycardia was more common in the dexmedetomidine group (14.2%) than in the midazolam group (5.2%; $p<0.001$). The two treatment groups showed no difference in neurocognitive adverse events after 48 hours of follow-up, including agitation, anxiety, and delirium (JAMA 2012;307:1151-60).
6. Adverse effects: Paradoxical agitation and prolonged duration of sedative action, respiratory depression, hypotension, bradycardia

I. Titration of Sedation in the ICU

1. Titration of medications using a sedation protocol to a goal level of sedation is arguably one of the most salient clinical practice standards within ICU care. This titration practice has consistently been shown to decrease time on mechanical ventilation, decrease ICU length of stay, and decrease rates of tracheostomy, which may all result in faster physical and cognitive rehabilitation time.
 - a. The goal level of sedation should be reassessed daily, documented, and communicated clearly to the nursing and medical staff because the goal may change throughout a patient's ICU stay.
 - b. Level of sedation should be assessed every 2–4 hours throughout the day and evening. Consider assessing every 4 hours during nighttime sleeping hours to minimize sleep interruption.
 - c. The two validated sedation scales currently recommended by the PADIS guidelines are the RASS (Table 5) and the Riker Sedation-Agitation Scale (Table 6).
2. Recommended methods for titration of medications include either (1) titration to a “light” versus “deep” level of sedation, unless clinically contraindicated, or (2) daily SAT. The PADIS guidelines suggest using light sedation (vs. deep sedation) in critically ill, mechanically ventilated adults. Light sedation is defined as a RASS of –2 to +1 or an SAS of 3 or 4, per the PADIS Guidelines. However, a pilot study from 2022 suggested that a RASS of –1 and an SAS of 4 indicate light sedation.
 - a. Patients should receive sedation only if required.
 - b. Sedatives should be titrated to allow patient responsiveness and awareness, as shown by patients' ability to purposefully respond to a combination of any three of the following actions on request: open eyes, maintain eye contact, squeeze hand, stick out tongue, and wiggle toes. A numerical score on a sedation scale may not fully represent a patient's ability to follow purposeful commands.
3. Data: In a multicenter, prospective, longitudinal Australian and New Zealand Study, Shehabi et al. evaluated 251 medical/surgical patients who were ventilated and sedated for 24 hours or more. Within 4 hours of initiating ventilation, deep sedation occurred in 191 patients (76.1%) and in 171 patients (68%) at 48 hours. Early deep sedation was an independent predictor of time to extubation (HR 0.90; 95% CI, 0.87–0.94; $p<0.001$), hospital death (HR 1.11, 95% CI, 1.02–1.20; $p=0.01$), and 180-day mortality (HR 1.08, 95% CI, 1.01–1.16; $p=0.026$) (JAMA 2012;307:1151-60).

4. In a single-center, nested cohort study within the Awakening and Breathing Controlled (ABC) randomized trial, Seymour et al. evaluated 140 patients for whom hourly doses of benzodiazepines and propofol during the daytime (7 a.m. to 11 a.m.) and nighttime (11 p.m. to 7 a.m.) for 5 days were measured. Greater daytime benzodiazepine doses were independently associated with failed SBT and extubation and subsequent delirium in adjusted models ($p < 0.02$ for all). A failed SBT ($p < 0.01$) and delirium ($p = 0.05$) were associated with nighttime increases in benzodiazepine doses (Crit Care Med 2012;40:2788-96).
- J. SAT Paired with SBT: The daily coordination of an SAT completed before an SBT is a method of weaning sedation before attempts at breathing trials in order to maximize a patient's chances of weaning from mechanical ventilation. This pairing of an SAT before an SBT is becoming recognized as an important component to ICU care and management of sedation. Important safety screens are incorporated into the daily SAT because studies have shown that the SAT is not appropriate for all ICU patients. If a patient does not pass the safety screen and does not undergo the SAT, this should not preclude the appropriate titration of sedatives to a goal level of sedation throughout the remainder of the day:
1. SAT safety screen (criteria may vary; published trial protocols have had variations): If any are present, discontinue the protocol and repeat in 12–24 hours or according to hospital protocol:
 - a. Current RASS greater than 2; or goal for deeper sedation (e.g., RASS -3 to -5)
 - b. Active seizures requiring a continuous infusion of a sedative to control
 - c. Active alcohol withdrawal requiring a continuous infusion of a sedative to control
 - d. Fio_2 of 70% or greater (these criteria are not consistently present among published trial protocols)
 - e. Neuromuscular blockade
 - f. Myocardial ischemia in previous 24 hours or ongoing myocardial ischemia
 - g. Intracranial pressure (ICP) greater than 20 mm Hg or need for control of ICP
 2. If pass SAT safety screen, begin SAT: Hold continuous sedative and analgesic infusions. Bolus opioids are recommended for breakthrough pain. Continuous opioid infusions allowed to continue while stopping sedatives if presence of active pain. If the patient “passes” the SAT, continue to the SBT safety screen.
 3. SAT failure (if any are present, discontinue the protocol, and repeat in 12–24 hours or according to hospital protocol):
 - a. Anxiety/agitation/pain present (e.g., RASS greater than +1 for 5 minutes or more)
 - b. Respiratory rate greater than 35 breaths/minute for 5 minutes or more
 - c. Oxygen saturation less than 88% for 5 minutes or more
 - d. ICP greater than 20 mm Hg
 - e. Acute cardiac ischemia or arrhythmia
 - f. Respiratory or cardiac distress (e.g., heart rate increase of 20 beats/minute or greater, heart rate less than 55 beats/minute, use of accessory muscles, abdominal paradox, diaphoresis, or dyspnea)
 4. If SAT fails: Consider giving patient bolus opioids first (up to three doses in 1 hour) before restarting infusion. Reinitiate sedation infusion, if necessary, at half the previous dose and titrate to goal. Determine the reasons for SAT failure. Repeat SAT steps in 12–24 hours or according to hospital protocol.
 5. SBT safety screen (if any are present, discontinue the protocol; repeat in 12–24 hours or according to hospital protocol):
 - a. Agitation
 - b. Oxygen saturation less than 88%, Fio_2 greater than 50%
 - c. PEEP (positive end-expiratory pressure) of 7 cm H_2O or greater
 - d. Myocardial ischemia in previous 24 hours
 - e. Increasing vasopressor requirements
 - f. Lack of inspiratory efforts

6. SBT: If a patient tolerates the SBT for 30 to 120 minutes, consider extubation.
 7. SBT failure:
 - a. Respiratory rate greater than 35 breaths/minute (for more than 5 minutes) or less than 8 breaths/minute
 - b. Oxygen saturation less than 88% for more than 5 minutes
 - c. ICP greater than 20 mm Hg, mental status change
 - d. Acute cardiac ischemia or arrhythmia
 - e. Respiratory distress (use of accessory muscles, abdominal paradox, diaphoresis, and dyspnea)
 8. If SBT fails: Place the patient on prior ventilator settings. Repeat bundle in 12–24 hours or according to hospital protocol.
- K. Sedation Protocol Compared with the Paired SAT-SBT Protocol: Studies comparing a standard sedation protocol with daily pairing of a SAT with SBT have shown decreased days on mechanical ventilation, days in the ICU, and decreased rates of delirium when the SAT is paired with the SBT.
1. The ABC trial included 336 mechanically ventilated patients from four tertiary care hospitals. Patients were randomized to patient-targeted sedation protocol plus the SBT (“usual care” control group) or to daily SAT paired with the SBT (intervention group). Both groups were deeply sedated on enrollment (RASS -4), and both groups had been admitted for 2.2 days before enrollment. In the intervention group, patients who passed the safety screen underwent an SAT: sedatives and analgesics used for sedation were discontinued, and analgesics used for active pain were continued. Patients “passed” their SAT if they opened their eyes to command or tolerated being off sedation for at least 4 hours without meeting failure criteria. The mean ventilator-free days was 11.6 days in the usual care control group versus 14.7 days in the SAT plus SBT group ($p=0.02$). The time to discharge was 12.9 days in the control group versus 9.1 days in the intervention group ($p=0.01$). Self-extubations were higher in the intervention group, but there was no difference in self-extubations requiring reintubation between groups. Rates of delirium assessed by the confusion assessment method for the intensive care unit (CAM-ICU) were no different between groups (74% vs. 71%).
 2. The first study evaluating the ABCDE (Awakening and Breathing Coordination, Delirium Monitoring/Management, and Early Mobility) Bundle compared clinical outcomes in patients before ($n=146$) and after ($n=150$) bundle-protocol implementation; 187 patients were on mechanical ventilation. The bundle protocol consisted of a daily-paired SAT/SBT, delirium screening with the CAM-ICU every 8 hours, and an early mobility protocol. The “before” bundle patients were enrolled from February to October 2011; the “after” bundle patients were enrolled from October 2011 to April 2012. There were some differences in patient type on admission, including more elective admissions in the post-bundle group (39 vs. 30), more cardiothoracic surgery patients in the post-bundle group (20 vs. 6), more surgical patients in the pre-bundle group (21 vs. 11), and more patients coming from an outside hospital in the pre-bundle group (9 vs. 1). The post-bundle group had more median ventilator-free days (24 vs. 21 days, $p=0.04$), less delirium at any time (49 vs. 62%, $p=0.03$), and shorter percentage of ICU days with delirium (33.3 vs. 50 %, $p=0.003$) (Crit Care Med 2014;42:1024-36).
 3. The SLEAP investigators from the Canadian Critical Care Trials Group studied the outcomes of patients receiving a daily sedation protocol alone versus patients receiving a daily sedation protocol plus a daily sedation interruption (Crit Care Med 2015;43:557-66; Crit Care Med 2015;43:2180-90; JAMA 2012;308:1985-92). From January 2008 to July 2011, 430 patients were enrolled from 16 tertiary care medical and surgical ICUs. Only opiate and benzodiazepine infusions were allowed in the study. According to the sedation-alone protocol, the RASS goal was -3 to 0, and the Sedation-Agitation Scale (SAS) goal was 3 or 4. Nurses assessed sedation levels on an hourly basis and titrated medications every 15–30 minutes to achieve sedation goals. If patients were oversedated in either group (SAS 1 or 2; RASS -4 or -5), infusions were discontinued. According to the sedation protocol with daily sedation

interruption, nurses stopped benzodiazepine and opiate infusions once a day and assessed hourly for wakefulness (e.g., a light SAS or RASS score, plus ability to follow at least three commands).

- a. Clinical outcomes (published 2012): There was no difference in the primary outcome of time to successful extubation between the two groups (7 days in both groups). There was a significant difference in time to extubation in the prespecified surgical/trauma group between sedation protocol with daily sedation interruption and sedation protocol alone (6 vs. 13 days; hazard ratio [HR] 2.55; 95% CI, 1.40–5.44). No difference in time to extubation was detected between groups among medical ICU patients (9 vs. 8 days; HR 0.92; 95% CI, 0.72–1.18). However, significantly lower daily doses of both benzodiazepines and opiates boluses and continuous infusions were used in the sedation protocol–alone group than in the sedation protocol plus interruption group. Although the sedation protocol suggested to target light sedation, the actual mean RASS/SAS was not reported for either group making it unclear how deeply sedated either group was.
- b. Delirium outcomes (published 2015): Delirium by the Intensive Care Delirium Screening Checklist (ICDSC) was diagnosed in 53.8% of patients in the study; there was no difference in delirium in the sedation protocol–alone group versus the protocol plus daily sedation interruption group. Patients who had delirium had a longer duration of mechanical ventilation, longer ICU and hospital stay, longer use of restraints, higher rates of tracheostomy, and higher incidence of unintentional device removal. Patients with delirium received almost the twice the mean dose of midazolam equivalents/patient/day (104 mg vs. 57 mg), higher fentanyl equivalents/patient/day (1497 mcg vs. 1150 mcg), more frequent use of anticholinergics (18 vs. 8.6%), and more frequent use of trazadone or zopiclone (17.7 vs. 9.8%) than did patients who were not delirious. Patients who developed delirium had a higher incidence of alcohol and cigarette use than did patients who did not develop delirium.
- c. Recall in ICU survivors (published 2015): The SLEAP investigator study did patient interviews on days 3, 28, and 90 post-ICU discharge to determine differences in recall between the sedation-alone protocol group and the protocol plus daily sedation interruption group. There were no differences in type of recall between the sedation strategies. Delusional memories were common at day 28 (70% of patients) but were unrelated to the presence of delirium or the total dose of benzodiazepines or opiates. Patients with no recall had received lower total doses of benzodiazepines than had patients with recall. Emotional memories such as panic and confusion declined over time.

Table 5. Richmond Agitation-Sedation Scale (RASS)

Score	Term	Description
+4	Combative	Overtly combative or violent; immediate danger to staff
+3	Very agitated	Pulls on or removes tube(s) or has aggressive behavior toward staff
+2	Agitated	Frequently nonpurposeful movement or patient-ventilator dyssynchrony
+1	Restless	Anxious or apprehensive, but movements not aggressive or vigorous
0	Alert and calm	
–1	Drowsy	Not fully alert but has sustained (> 10 s) awakening, with eye contact, to voice
–2	Light sedation	Briefly (< 10 s) awakens with eye contact to voice
–3	Moderate sedation	Any movement (but no eye contact) to voice
–4	Deep sedation	No response to voice, but any movement to physical stimulation
–5	Unarousable	No response to voice or physical stimulation

Table 5. Richmond Agitation-Sedation Scale (RASS) (*continued*)

Score	Term	Description
Procedure		
1.	Observe patient. Is patient alert and calm (score 0)? Does patient have behavior that is consistent with restlessness or agitation (score +1 to +4 using the criteria listed above, under <i>Description</i>)?	
2.	If patient is not alert, in a loud speaking voice, state the patient's name and direct the patient to open eyes and look at speaker. Repeat once if necessary. Can prompt patient to continue looking at speaker. Patient has eye opening and eye contact, which is sustained for more than 10 seconds (score –1). Patient has eye opening and eye contact, but this is not sustained for 10 seconds (score –2). Patient has any movement in response to voice, excluding eye contact (score –3).	
3.	If patient does not respond to voice, physically stimulate patient by shaking shoulder and then rubbing sternum if there is no response to shaking shoulder. Patient has any movement to physical stimulation (score –4). Patient has no response to voice or physical stimulation (score –5).	

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Table 6. Riker Sedation-Agitation Scale

7	Dangerous agitation	Pulling at ETT, trying to remove catheters, climbing over bedrail, striking at staff, thrashing side to side
6	Very agitated	Does not calm despite frequent verbal reminding of limits, requires physical restraints, biting ETT
5	Agitated	Anxious or mildly agitated, trying to sit up, calms down to verbal instructions
4	Calm and cooperative	Calm, awakens easily, follows commands
3	Sedated	Difficult to arouse, awakens to verbal stimuli or gentle shaking but drifts off again, follows simple commands
2	Very sedated	Arouses to physical stimuli but does not communicate or follow commands, may move spontaneously
1	Unarousable	Minimal or no response to noxious stimuli, does not communicate or follow commands

ETT = endotracheal tube.

Adapted with copyright permission from: Lippincott Williams and Wilkins/Wolters Kluwer Health. Simmons LE, Riker RR, Prato BS, et al. Assessing sedation during intensive care unit mechanical ventilation with the Bispectral Index and the Sedation Agitation Scale. *Crit Care Med* 1999;27:1499-504.

L. Acute Withdrawal Syndrome of Long-term Analgesia and/or Sedation in the ICU

1. Patients who have been receiving high doses of continuous infusion sedation and/or analgesia in the ICU for an extended period may be at risk of sedative or analgesia withdrawal as dose tapering begins. In a retrospective review of adult trauma/surgical ICU patients, 32% of patients experienced either sedative or opiate withdrawal soon after discontinuing these medications. The patients in this study had been in the ICU for 20 days or more and were receiving higher mean daily analgesic and sedative doses than were the non-withdrawal patients (fentanyl 6.4 mg vs. 1.4 mg; lorazepam 38 mg vs. 11 mg). The withdrawal patients in this study were also more likely to have received an NMBA (*Crit Care Med* 1998;26:676-84).
2. Data: An international, multicenter, observational, point prevalence study in 229 adult ICUs showed that a small number of ICUs have policies or protocols for opioid or sedative weaning or iatrogenic withdrawal syndrome. Even when these protocols exist, they are rarely used.

3. The risk factors and incidence of sedation or analgesia withdrawal in adult ICU patients have not been well characterized; however, these are important considerations in the long-term ICU patient receiving high doses of these medications. Use of longer-acting agents given orally or by feeding tube has been described to assist in the transition off long-term continuous infusions. The medical indication and dosing plan for using oral medications to taper off continuous infusions should be clearly documented in the medical chart on patient discharge from the ICU. Clonidine is a potential consideration in patients to aid in dexmedetomidine withdrawal.

Patient Cases

3. A 48-year-old man with cirrhosis and now hepatorenal syndrome was intubated for respiratory distress. He has been receiving midazolam 1 mg/hour and fentanyl 75 mcg/hour for 2 days; his RASS (-4 to -5) and CPOT has been 1 for 24 hours. Oxygen requirements have decreased, and vital signs are normal. Which is the most appropriate change in his medications?
 - A. Decrease midazolam; give as-needed lorazepam for agitated RASS score.
 - B. Discontinue midazolam and fentanyl; give as-needed fentanyl for CPOT of 3 or greater.
 - C. Discontinue midazolam; initiate propofol drip.
 - D. Change midazolam to dexmedetomidine drip.
4. T.L. is a 55-year-old woman intubated for respiratory distress for severe pneumonia. She is receiving fentanyl 50 mcg/hour and dexmedetomidine 1.0 mcg/kg/hour. Her home medications are confirmed to include esomeprazole 20 mg daily, lorazepam 1 mg three times daily, and citalopram 10 mg daily. The nurse reports intermittent agitation with tachycardia and a negative pain score. Which is the most appropriate recommendation?
 - A. Increase fentanyl drip for agitated RASS score.
 - B. Reinitiate lorazepam and citalopram.
 - C. Give fentanyl boluses as needed for agitation.
 - D. Increase dexmedetomidine.

IV. DELIRIUM IN THE INTENSIVE CARE UNIT

- A. Delirium is an acute and fluctuating disturbance in consciousness resulting in the inability to receive, process, store, or recall information. In the ICU, delirium may present as hyperactive (agitated and restless), hypoactive (flat affect, apathy, lethargy, decreased responsiveness), or mixed hyper/hypoactive states. Most common in the ICU are mixed and hypoactive states of delirium. Two screening tools are currently recommended by the PAD guidelines: (1) the CAM-ICU and (2) the ICDSC. Both the CAM-ICU and the ICDSC require a RASS (-2) or a SAS (3) or more alert to be completed.
 1. The CAM-ICU assesses four features: (1) acute change or fluctuation in mental status from baseline, (2) inattention, (3) altered level of consciousness, and (4) disorganized thinking. If features 1 *and* 2 plus feature 3 *or* 4 are present, the patient is considered positive for delirium. Detailed training is available at www.icudelirium.org.

2. The ICDSC consists of eight items, evaluated during an 8- to 24-hour period. The eight symptoms are level of consciousness, inattention, disorientation, hallucinations-delusions-psychosis, psychomotor agitation or retardation, inappropriate speech or mood, sleep-wake cycle disturbances, and symptom fluctuation. A point is given for any symptom that is present during the previous 24 hours; a score of 4 or higher indicates the presence of delirium.
- B. Background – 30%–80% of ICU patients reportedly develop delirium, depending on the severity of illness and the diagnostic method, yet assessment for delirium is still not routine in most U.S. ICUs. During a patient's hospitalization, the presence of delirium is associated with difficulty in weaning mechanical ventilation and longer duration of mechanical ventilation, increased use of physical and chemical restraints, longer duration of ICU stay, and additional stress to family and friends who may not understand the course of delirium. Delirium is also associated with up to a 3-fold increase in mortality, increase in cognitive decline, delay in cognitive recovery, and increased likelihood of being discharged to a nursing home. Two studies found that a longer duration of delirium was independently associated with worse activity of daily living scores and worse cognitive impairment scores at 3 and 12 months post-ICU discharge (Crit Care Med 2014;42:369-77; N Engl J Med 2013;369:1306-16). A retrospective study reported increased difficulty in the weaning of mechanical ventilation when delirium was detected in patients during the first spontaneous weaning trial compared with in patients who did not have delirium (Respirology Nov 2015;1-8).
1. The underlying pathophysiology of delirium is not well understood; however, it may involve a complex set of factors:
 - a. Cerebral hypoperfusion and alterations in cerebral blood flow
 - b. Degradation of the blood-brain barrier, causing influx of inflammatory cytokines and microvascular thrombosis
 - c. Depletion in central neurotransmitters (e.g., dopamine, norepinephrine, serotonin)
 - d. Depletion in acetylcholine
 - e. Medication withdrawal
 2. Risk factors for delirium: A systematic review of studies from 2001 to 2013 described 11 variables identified as risk factors for developing delirium in the ICU, extracting from only a strong or moderate level of evidence (Crit Care Med 2015;43:40-7):
 - a. Age
 - b. Preexisting dementia
 - c. History of baseline hypertension
 - d. Sedative-associated coma
 - e. APACHE II (Acute Physiology and Chronic Health Evaluation II) score
 - f. Delirium on the previous day
 - g. Emergency surgery
 - h. Mechanical ventilation
 - i. Organ failure
 - j. (Poly)trauma
 - k. Metabolic acidosis
 3. Other reported risk factors or precipitants:
 - a. Infection
 - b. Dehydration or malnutrition
 - c. Sleep deprivation
 - d. Centrally acting medications (benzodiazepines, opiates, anticholinergics)
 - e. Lack of exposure to sunlight
 - f. Lack of personal interaction
 - g. Physical restraints or insertion of catheters or tubes

4. Medication-induced altered mental status – Although the development of delirium is considered multifactorial, any patient who presents with a change in mental status should have his or her medications and doses immediately scrutinized as part of the initial workup for delirium. Several classes of medications have long been recognized for their effects on mental status and cognitive function, in or out of the ICU. These medications have the potential to affect a patient's level of consciousness or course of delirium at any point in the patient's hospital stay. Anticholinergics, benzodiazepines, opiates, antipsychotics, antispasmodics, anticonvulsants, corticosteroids, and others should be used with caution in a hospitalized patient, with close monitoring of the patient's cognitive adverse effects. Because renal and hepatic function may fluctuate throughout an ICU stay and affect the clearance of these medications, doses must be thoughtfully titrated. Research on the degree of impact these medications have on the overall course of sedation and delirium in the ICU is difficult to characterize, and research is undergoing. Given the multifactorial nature of delirium in the ICU, clinicians should be leery of solely assigning blame to medications but should remain vigilant when assessing the need and doses of the aforementioned medications.
 - a. Benzodiazepines: The PADIS guidelines state "benzodiazepine use may be a risk factor for the development of delirium in adult ICU patients." Of interest, research directed at finding an independent association between the use of benzodiazepines and the development of delirium in the ICU including a meta-analysis, a randomized trial, and a systematic review has yielded mixed results (Crit Care Med 2015;43:40-7; Crit Care Med 2015;43:557-66; Crit Care Med 2013;41:S30-8). Investigators used more appropriate statistical analysis for a fluctuating illness such as delirium (e.g., Markov monitoring was used to determine the probability of a daily transition to delirium while assuming this was independent of the patient history beyond the prior day) in addition to more frequent delirium monitoring to examine the use of benzodiazepines and the transition from an awake state without delirium to delirium, or from coma to delirium by the next day (Intensive Care Med 2015;41:2130-7). This study found that a midazolam equivalent dose of just 5 mg/day increased the odds of developing delirium the next day by 4%, and the use of benzodiazepine infusions was an independent risk factor for the transition to delirium in the study population. However, there was a large difference in the daily dose of benzodiazepines between the continuous infusion (daily median dose 99 mg) compared with the intermittent dosing group (daily median dose 4.1 mg), bringing into question whether the risk factor for the transition to delirium was indeed the cumulative dose and not the method of administration. As the data evolve, the scrutiny of benzodiazepine use should persist for ICU clinicians. Routine strategies to preferentially use a non-benzodiazepine sedative (e.g., propofol or dexmedetomidine) and to avoid benzodiazepine infusions unless clinically indicated should be used in addition to diligent performance of daily SATs to discontinue use as soon as possible.
 - b. Anticholinergics: These medications are known for their sedating and altering effects on mentation and should be avoided or used with extreme caution in the ICU setting. One proposed mechanism for delirium is a decline in acetylcholine concentrations; therefore, any medication that may further inhibit the activity of acetylcholine could worsen the patient's mental status. A prospective cohort study of 1112 critically ill patients was conducted to determine whether anticholinergic exposure increased the probability of transitioning to delirium. On 6% of ICU days, transition from "awake and without delirium" to "delirium" occurred. A nonsignificant increase in the probability of transitioning to delirium the following day resulted from a 1-unit increase in the Anticholinergic Drug Scale (odds ratio 1.05; 95% CI, 0.99–1.10). The dose of the medication was not evaluated to determine the effects on the transition to delirium, and it was not evaluated if patients were already delirious and remained delirious while receiving anticholinergic medications (Crit Care Med 2015;43:1846-52).

- c. Systemic corticosteroids: The neuropsychiatric effects of systemic steroids in various clinical settings have been described for more than 50 years. Common symptoms include mania, depression, mood lability, anxiety, insomnia, delirium, and psychosis. The incidence of these symptoms varies greatly depending on the clinical setting, dose of steroid, and patient's underlying medical history. Reported risk factors for neuropsychiatric effects secondary to steroids include a daily prednisone dose equivalent to 40 mg or greater, hypoalbuminemia, underlying psychiatric disorder, and blood-brain barrier damage (Curr Opin Organ Transplant 2014;19:201-8). Although steroids are commonly used in the ICU for various indications and at various doses, significant research directed at the neuropsychiatric effects of steroids in the critically ill population is lacking. In a secondary analysis of a multicenter observational study of adult medical and surgical ICU patients with acute lung injury (n=330), Schreiber et al. found a significant and independent association between the use of systemic corticosteroids and the transition to delirium from a non-comatose, non-delirious state within 24 hours of corticosteroid administration (odds ratio [OR] 1.52 [1.05–2.21], p=0.03). Delirium was documented on one or more days in 83% of patients, with a median duration of 7 days. There was no significant association in prednisone-equivalent dose and transition to delirium. Schreiber et al. recognize that a direct causal relationship could not be determined between corticosteroid use and delirium from this observational study; however, they believe that the study adds valuable data toward our understanding of risk factors for delirium in the ICU (Crit Care Med 2014;42:1480-6). A second study that investigated steroids and transition to delirium in a mixed medical and surgical ICU population (n=1112) found no association between steroid use and a transition to delirium. The median prednisone equivalent dose was 50 mg (Crit Care Med 2015;43:e585-8).
 5. Outcomes of sedation-related versus illness-related delirium: A single-center study using propofol and fentanyl timed its CAM-ICU assessments before and after a daily sedation interruption protocol. Rapidly reversible delirium was defined as delirium while patients were receiving sedation that resolved within 2 hours after performing an SAT. This type of delirium was rare (12% of the 102 patients), but these patients has a prognosis that was similar to patients who did not have delirium. Most patients (75%) had persistent delirium, delirium that did not resolve with cessation of sedatives, a higher risk of death, and longer length of stay (Am J Respir Crit Care Med 2014;189:658-65). Patients can have both sedation- and illness-related delirium, and additional research in this area is needed to clarify the differences in short- and long-term outcomes.
- C. Monitoring for Delirium: SCCM recommends assessing critically ill patients for delirium with a validated tool such as either the CAM-ICU or the ICDSC. The PAD guidelines summarized their review of five delirium assessment scales used for adult ICU patients. The two scales with the highest psychometric (e.g., validity and reliability) scores were the CAM-ICU and the ICDSC. Both scales were designed for patients in the ICU either on or off mechanical ventilation, and both showed high sensitivity and specificity when tested against the American Psychiatric Association's criteria for delirium.
1. Delirium should be assessed at least every 8-12 hours and documented in the medical chart; results should be discussed with the medical team. Because these assessment scales cannot distinguish between sedation- and disease-related causes of delirium, delirium assessments should ideally be timed both before and after SATs with appropriate time allowed for drug clearance (www.icudelirium.org, Am J Respir Crit Care Med 2014;189:658-65). If this timing is not feasible and a patient screens positive for delirium while receiving ongoing analgesia or sedation, an SAT should be conducted if the patient passes the safety screen to assist in ruling out a medication-induced cause of delirium.

2. If a patient's delirium score is positive, the medical team should correct possible etiologies (e.g., decrease sedative doses, if safe), decrease ongoing risk factors, address inciting factors (e.g., metabolic derangements, infection, withdrawal), and try nonpharmacologic treatment and preventive measures when appropriate.
- D. Prevention of Delirium: With a lack of data supporting the use of pharmacologic agents to prevent delirium, the PADIS guidelines suggest not using haloperidol, an atypical antipsychotic, dexmedetomidine, a statin, or ketamine to prevent delirium in critically ill patients. Instead, the recommendations are focused on nonpharmacologic prevention methods when feasible, particularly for patients at high risk of delirium. Preventive efforts may help avert 30%–40% of new-onset delirium cases, particularly in older adults. Recommended nonpharmacologic strategies by the PADIS guidelines include:
1. Performing rehabilitation or mobilization in critically ill adults
 2. Using a multicomponent intervention to reduce or shorten delirium (e.g., reorientation, cognitive stimulation, use of clocks); improve sleep (e.g., minimize light and noise); improve wakefulness (i.e., reduce sedation); reduce immobility (e.g., early rehabilitation/mobilization); and reduce hearing and/or visual impairment (e.g., enable use of devices such as hearing aids or eyeglasses).
 3. Decreasing the use of benzodiazepines and anticholinergics in patients at risk of delirium; use the lowest effective doses of any sedating medication (e.g., opiates, antipsychotics).
 4. REDUCE was a randomized, double-blind, placebo-controlled study of 1789 critically ill patients randomized to prophylactic haloperidol 1 mg, haloperidol 2 mg, or placebo. The 1-mg haloperidol group was prematurely stopped because of futility (JAMA 2018;319:680-90). No difference occurred in the median days that patients survived in 28 days in the 2-mg haloperidol group compared with 28 days in the placebo group (95% CI, 0-0; $p=0.93$) with a hazard ratio of 1.003 (95% CI, 0.78–1.30; $p=0.82$). None of the 15 secondary outcomes were statistically different between the three groups. These outcomes included delirium incidence (mean difference 1.5%; 95% CI, –3.6% to 6.7%), delirium- and coma-free days (mean difference 0 days; 95% CI, 0–0 days), and duration of mechanical ventilation, ICU, and hospital length of stay (mean difference 0 days; 95% CI, 0–0 days for all three measures). Adverse events did not differ between groups.
- E. Sleep in the ICU: Uninterrupted sleep (ideally 4 hours or more) is vital for an adequate immune response to illness, to maintain normal metabolic and hormonal balance, and to help decrease delirium and/or agitation. Disturbances in the ICU such as multiple alarms and frequent physical interruptions (e.g., examination, turning, laboratory tests, medication administration) make it challenging for patients to maintain the slow-wave sleep cycle needed for optimal immune function. Sleep research in the ICU is ongoing, and more information will be forthcoming regarding its effects in the critically ill patient. The PADIS guidelines suggest not routinely using physiologic sleep monitoring clinically in critically ill adults. In addition, the PADIS guidelines suggest not using aromatherapy, acupuncture, or music at night to improve sleep in critically ill adults. However the PADIS guidelines do suggest using noise and light reduction strategies to improve sleep in critically ill adults. The PADIS guidelines make no recommendation regarding the use of melatonin to improve sleep or regarding the use of dexmedetomidine at night to improve sleep. The PADIS guidelines suggest not using propofol to improve sleep in critically ill adults. In addition, the PADIS guidelines suggest using a sleep-promoting, multicomponent protocol in critically ill adults.
1. To avoid waking patients at night, pharmacists should ensure that medications are scheduled during the daytime and evening hours, if possible—particularly orally or subcutaneously administered medications.
 2. Sleep protocols should seek to cluster patient care activities (e.g., vital sign checks, radiology tests, laboratory checks, sedation assessments) around nighttime sleeping hours unless clinically indicated in a specific patient population.

3. A two-center, double-blind randomized trial randomized 100 critically ill adults without delirium to nocturnal dexmedetomidine or placebo (Am J Respir Crit Care Med 2018;197:1147-56). During the ICU stay, nocturnal dexmedetomidine was associated with a greater proportion of patients who remained delirium free (dexmedetomidine (40 of 50 patients [80%]) than with placebo (27 of 50 patients [54%]; relative risk 0.44; 95% CI, 0.23–0.82; $p=0.006$). Adverse events did not differ between the two groups.
- F. Treatment of Delirium: The cause of delirium may be multifactorial, and identifying and correcting the underlying etiology is the first step in management. Patients can also progress to alcohol withdrawal or withdrawal from other chronic medications/substances and present with hyperactive delirium. The PADIS guidelines suggest not using an atypical antipsychotic, haloperidol, or a statin to treat subsyndromal delirium or delirium (N Engl J Med 2018;379:2506-16). The Modifying the Impact of the ICU-Associated Neurological Dysfunction-USA (MIND USA) Study is a multicenter, randomized, placebo-controlled study of 566 patients showing that haloperidol and ziprasidone did not reduce delirium, time on the ventilator, ICU or hospital length of stay, or death compared with placebo. Arrhythmias, parkinsonism (extrapyramidal symptoms), neuroleptic malignant syndrome, study drug discontinuation, and other safety concerns were extremely low across all three groups. Additionally, a planned secondary analysis of the MIND USA study showed that neither haloperidol nor ziprasidone significantly increased QTc intervals compared with placebo. The AID-ICU trial was a multicenter, randomized, placebo-controlled trial of 963 patients with delirium randomized to haloperidol or placebo. The mean number of days alive and out of the hospital was 35.8 (95% CI, 32.9–38.6) in the haloperidol group and 32.9 (95% CI, 29.9–35.8) in the placebo group with an adjusted mean difference of 2.9 days (95% CI, -1.2 to 7; $p=0.22$). Antipsychotics remain viable for the short-term control of agitation (e.g., alcohol or drug withdrawal) or severe anxiety with the need to avoid respiratory suppression (e.g., heart failure, COPD, or asthma). If an antipsychotic is initiated, low starting doses should be considered, and daily review of drug interactions, adverse effects, dosing titration, and need for the antipsychotic should be completed. In addition, a strategy for discontinuation or outpatient follow-up should be documented to help avoid inadvertent continuation beyond the hospital environment (Table 7). Serious adverse effects are associated with the use of any antipsychotic; effects such as arrhythmias, serotonin syndrome, neuroleptic malignant syndrome, extrapyramidal symptoms, and oversedation should be closely monitored on a daily basis. Dose ranges for atypical antipsychotics for ICU delirium are not well described. The American Geriatrics Society 2015 Beers Criteria for medication use in older adults includes the following recommendation: “Avoid antipsychotics for behavioral problems of dementia or delirium unless nonpharmacological options have failed or are not possible AND the older adult is threatening substantial harm to self or others.” If the ICU team decides to use antipsychotics in older adults, lower starting doses should be considered, together with daily review of drug interactions and adverse effects. The PADIS guidelines suggest using dexmedetomidine for delirium in mechanically ventilated adults when agitation precludes weaning/extubation. The PADIS guidelines suggest not using bright light therapy to reduce delirium in critically ill adults.
- G. Inadvertent Continuation of Antipsychotics Beyond ICU Discharge: The PADIS guidelines discuss the use of adjunctive medications for ICU patients (e.g., antipsychotics, gabapentin, carbamazepine). Although their use in the ICU may be appropriate, there is a potential for inadvertent continuation of these medications on hospital discharge if a treatment plan is not clear in the medical record. This has been a well-documented problem with other medications initiated in the ICU (e.g., histamine receptor blockers, proton pump inhibitors), and studies have been published describing the continuation of newly prescribed antipsychotics from the ICU and hospital, even when an indication for continuation was not documented (J Crit Care 2015;30:814-6; J Crit Care 2016;33:119-24). Continued use of these medications beyond the hospital stay could lead to serious adverse effects, drug interactions, and significant drug cost as well as a presumption of a psychiatric or neuromuscular disorder associated with these drugs. Communication to the next direct patient care provider is crucial to appropriately direct the next steps in medication reconciliation.

Table 7. Antipsychotics^a

Drug	CYP Substrate (major)	Usual Starting Dose	Significant Adverse Effects ^b	Formulations
Haloperidol	3A4, 2D6	1–2 mg older adults; 2–4 mg if history of psychiatric disorders	Anticholinergic: * Sedation: * EPS: ** NMS: *	PO, IM, IV
Olanzapine	1A2	5 mg	Anticholinergic: ** Sedation: ** EPS: * NMS: * Neuromuscular weakness	PO, disintegrating tablet, IV, IM
Quetiapine	3A4	12.5–25 mg	Anticholinergic: ** Sedation: ** NMS: * Orthostatic hypotension: **	PO
Risperidone	2D6	0.5–1 mg	Anticholinergic: * Sedation: * EPS: ** NMS: * Orthostatic hypotension: ** Cardiac conduction abnormalities	PO, disintegrating tablet
Ziprasidone	1A2 (minor) 3A4 (minor)	20 mg PO; 10 mg IM	Anticholinergic: * Sedation: * EPS: * NMS: *	Oral, IM

NOTE: * = lower risk; ** = medium-higher risk

^aNot all medications listed are FDA label approved for use in delirium; not all are recommended by SCCM for the treatment of delirium in the ICU.^bAdverse effects other than QTc prolongation. Documented QTc prolongation incidence: IV haloperidol = ziprasidone > risperidone > olanzapine = quetiapine.

EPS = extrapyramidal symptoms; IM = intramuscular(ly); IV = intravenous(ly); NMS = neuroleptic malignant syndrome; PO = oral(ly)

1. Quetiapine (Seroquel): A randomized, placebo-controlled pilot trial compared the efficacy and safety of scheduled quetiapine with placebo for the treatment of delirium in ICU patients during a 10-day study (Crit Care Med 2010;38:419-27). Significant exclusions were as follows: patients with end-stage liver disease, those with alcohol withdrawal, those with a QTc greater than 500, and those receiving concomitant QTc-prolonging agents. This small pilot study (n=36), in which the placebo group was administered as-needed intravenous haloperidol, found that quetiapine was associated with a shorter time to first resolution of delirium, reduced duration of delirium, and less agitation than placebo. Mortality and ICU length of stay were not different from placebo.
 - a. Pharmacokinetics: Hepatically metabolized to one active and two inactive metabolites. Metabolites renally cleared. Many drug interactions, CYP3A4 (major) and CYP2D6 (minor) substrates. Peak plasma concentrations for oral about 1½ hours (immediate release).
 - b. Initial dose range for ICU delirium: 50 mg one to three times daily. Consider lower starting doses for older adult patients because of sedating effects. The Devlin study initiated 50 mg every 12 hours and titrated to a maximum dose of 200 mg every 12 hours.
 - c. Adverse effects (early onset): Sedation, orthostatic hypotension, extrapyramidal symptoms, QTc prolongation

2. Olanzapine (Zyprexa): Available in oral, orally disintegrating, and intramuscular (immediate and extended release) dosage forms. Intramuscular administration may result in plasma concentrations 5 times those of oral administration. The immediate release intramuscular formulation can also be administered intravenously. The U.S. Food and Drug Administration (FDA) warns that the use of intramuscular olanzapine has resulted in unexplained deaths; use of intramuscular olanzapine with benzodiazepines may result in significant oxygen desaturation.
 - a. Pharmacokinetics: Metabolized by glucuronidation and CYP 1A2, 2D6 oxidation. Clearance is significantly increased (around 40%) in smokers and decreased in females (around 30%). Many drug interactions, CYP1A2 (major) and CYP2D6 (minor) substrates. Weak inhibitor of several CYP isoenzymes. Peak plasma concentrations for oral: About 6 hours.
 - b. Suggested starting dose for ICU delirium: 5 mg orally once daily
 - c. Adverse effects (early onset): Drowsiness, extrapyramidal symptoms, neuromuscular weakness, serotonin syndrome. High doses may cause cardiac arrhythmias, cardiopulmonary arrest, and extreme sedation to coma-like states.
3. Risperidone (Risperdal): Available in oral and oral dispersible tablets (M-tabs) and intramuscular injection dosage forms
 - a. Pharmacokinetics: Hepatically metabolized to active metabolites, renally cleared. Many drug interactions, CYP2D6 (major) and CYP3A4 (minor) substrates and P-glycoprotein. Peak plasma concentrations for oral about 1 hour.
 - b. Suggested starting dose for ICU delirium: 0.25–0.5 mg once or twice daily
 - c. Adverse effects (early onset): Cardiac arrhythmias, anticholinergic effects, extrapyramidal symptoms
4. Ziprasidone (Geodon): Studied in a multicenter, randomized, placebo-controlled pilot trial of mechanically ventilated patients to test the hypothesis that antipsychotics would improve days alive without delirium or coma in the ICU (MIND trial). Medical and surgical adult ICU patients (n=101) from six tertiary-care centers in the United States on mechanical ventilation who had an abnormal level of consciousness or were receiving analgesia/sedative medications were randomly assigned to receive haloperidol, ziprasidone, or placebo every 6 hours for up to 14 days during a 21-day study. During the study, no difference was found in median days alive without delirium or coma between the haloperidol (14 days), ziprasidone (15 days), and placebo (12.5 days) groups, $p=0.66$. The study also found no difference in ventilator-free days, hospital length of stay, or mortality among the three groups (Crit Care Med 2010;38:428-37).
 - a. Ziprasidone is available in oral and intramuscular dosage forms.
 - b. Pharmacokinetics: Hepatic by glutathione and aldehyde oxidase. Minor substrates of CYP 1A2, 3A4. Peak plasma concentrations for oral about 6 hours; intramuscular about 1 hour.
 - c. Suggested starting dose for ICU delirium: 20 mg twice daily (oral)
 - d. Adverse effects (early onset): Somnolence, extrapyramidal symptoms, dizziness, orthostatic hypotension

V. ABCDEF BUNDLE

- A. Incorporating multiple concomitant patient care interventions into one consolidated bundle may be an effective strategy to improve clinical outcomes in critically ill patients. SCCM recommends implementing the “ABCDEF” bundle to align and coordinate care using an interprofessional approach (e.g., physician, nursing, pharmacy, respiratory therapy, physical and occupational therapy). The following practice principles are applied to the bundle:
1. A: Assess, prevent, and manage pain
 2. B: Both SATs and SBTs
 3. C: Choice of analgesia and sedation
 4. D: Delirium: Assess, prevent, and manage
 5. E: Early mobility and exercise
 6. F: Family engagement and empowerment.
- B. Assess, Prevent, and Manage Pain (“A” of the bundle): Asking the patient to self-report pain or to use the CPOT or BPS if the patient is nonverbal. Preventing pain by recognizing patients with known sources of pain (i.e., rib fractures) and scheduling analgesics when indicated. Managing pain by ordering the most appropriate pharmacologic agent on the basis of the source of pain and renal and liver function.
- C. Daily coordination of the SAT with the SBT (“B” of the bundle) versus usual care with the SBT has been shown to significantly decrease the time on mechanical ventilation and ICU length of stay in randomized studies. This was reviewed earlier in the chapter in the Agitation section.
- D. Choice of Sedation (“C” of the bundle): Use a multidisciplinary approach, including focused pharmacy input, to choose a sedative according to individual patient needs, hemodynamic stability, and organ function (e.g., hepatic, renal, cardiac, pulmonary, pancreatic).
- E. Delirium Assessment, Prevention, and Management (“D” of the bundle): Regularly assess for delirium using the CAM-ICU or the ICDSC every 8–12 hours. Use delirium preventive measures in all patients when safe to do so.
- F. Early Mobility (“E” of the bundle): Perform a mobility safety screen, and implement a daily mobility protocol.
- G. Family Engagement and Empowerment (“F” of the bundle): Good communication with the family is critical at every step of the patient’s clinical course, and empowering the family to be part of the team to ensure best care is adhered to diligently will improve many aspects of the patient’s experience.
- H. In a prospective, multicenter, cohort study from 68 academic, community, and federal ICUs during a 20-month collection period, performance of the complete ABCDEF bundle was associated with a lower likelihood of death within 7 days (HR 0.32; CI, 0.17–0.62), next-day mechanical ventilation (OR 0.28; CI, 0.22–0.36), coma (OR 0.35; CI, 0.22–0.56), delirium (OR 0.60; CI, 0.49–0.72), physical restraint use (OR 0.37; CI, 0.30–0.46), ICU readmission (OR 0.54; CI, 0.37–0.79), and discharge to a facility other than home (OR 0.64; CI, 0.51–0.80). There was a dose response in between higher proportional bundle performance and improvement in each clinical outcome ($p<0.002$). Pain was more commonly reported as bundle performance increased ($p=0.0001$), probably because more patients were awake.

VI. POST-INTENSIVE CARE SYNDROME

- A. Advances in critical care have decreased mortality and resulted in an increased likelihood of surviving critical illness. Post-intensive care syndrome (PICS) describes new or worsening impairments in physical, cognitive, or mental health status after critical illness and persisting beyond acute care hospitalization. Physical impairments include both pulmonary dysfunction and neuromuscular weakness. Impairments in memory and executive functioning are examples of cognitive dysfunction. Mental health impairments include depression, posttraumatic stress disorder, and anxiety. Medication management including glucose control, use of the ABCDEF bundle, and review of medication lists at every transition of care are important roles of the pharmacist to prevent PICS. Family members can also have PICS, termed PICS-F, in which these individuals have depression, posttraumatic stress disorder, anxiety, and prolonged grief. SCCM has taken progressive steps to help clinicians and families recognize the prolonged PICS and PICS-F through THRIVE. The establishment of an ICU follow-up clinic is one proposed method to manage long-term complications of patients with PICS and family members of PICS-F through optimization of physical, cognitive, and mental health; improved coordination of care; and reduction in health care use. The pharmacist should be considered a key member of the PICS clinic team who performs complete medication management on all patients seen in the clinic.
1. Physical impairment: A prospective, longitudinal study of 109 survivors with ARDS was conducted. Median (Interquartile Range) Total Lung Capacity (TVC) was 92% (77–97%), 92% (82–101%), and 95% (81–103%) of predicted value, respectively, at 1, 6, and 12 months after ICU discharge. Forced expiratory volume in 1 second (FEV₁) was 75% (58–92%), 85% (69–98%), and 86% (74–100%) of predicted value, respectively, at 1, 6, and 12 months after ICU discharge. Six-minute walk test was 49%, 64%, and 66% of predicted value, respectively, at 1, 6, and 12 months after ICU discharge. These 109 survivors of ARDS were further analyzed annually up to 5 years after ICU discharge. TVC was 94% (84–108%), 93% (78–107%), 92% (79–104%), and 94% (78–105%) of predicted value, respectively, at 2, 3, 4, and 5 years after ICU discharge. FEV₁ was 87% (75–99%), 79% (66–97%), 85% (68–98%), and 83% (69–98%) of predicted value, respectively, at 2, 3, 4, and 5 years after ICU discharge. Six-minute walk test was 68%, 67%, 71%, and 76% of predicted value, respectively, at 2, 3, 4, and 5 years after ICU discharge.
 2. Cognitive impairment: A large, multicenter, prospective observational cohort study of 821 adult medical ICU and surgical ICU patients (called the BRAIN-ICU study) estimated the prevalence of long-term cognitive impairment after critical illness secondary to respiratory failure, cardiogenic shock, or septic shock (N Engl J Med 2013;369:1306–16). Delirium was the strongest independent predictor of cognitive impairment in the 50% of patients after critical illness. A Repeatable Battery for Neuropsychological Status (RBANS) score similar to Alzheimer disease (2 standard deviations below the population mean) was found in 26% of patients, and a score similar to moderate traumatic brain injury (1.5 standard deviations below the population mean) was found in 40% of patients 3 months after discharge. Both young and older adults, with and without comorbidities, experienced these deficits, which persisted at 12 months in 24% and 34% of these individuals having RBANS scores similar to Alzheimer disease and moderate traumatic brain injury, respectively.
 3. Mental health impairments: A multicenter, prospective observational cohort study of 821 adult medical ICU and surgical ICU patients estimate the prevalence of depression, posttraumatic stress disorder (PTSD), and functional disability after critical illness secondary to respiratory failure, cardiogenic shock, or septic shock. Depression was found in 149 patients (37%) and 116 patients (33%) at 3 and 6 months after ICU discharge, respectively. Disabilities in basic activities of daily living were found in 139 patients (32%) and 102 patients (23%) at 3 and 6 months after ICU discharge, respectively. Disabilities in instrumental activities of daily living were found in 108 patients (26%) and 87 patients (23%) at 3 and 6 months after ICU discharge, respectively. PTSD was found in 27 patients (7%) at 3 and 6 months after ICU discharge.

4. A prospective, observational feasibility study was conducted at an academic hospital between July 2012 and December 2015 (Crit Care 2018;46:141-8). Patients were identified if at high risk of PICS; if so, post-ICU care was offered. Sixty-two patients were seen in the clinic. Median time from hospital discharge to ICU recovery center visit was 29 days. Cognitive impairment was identified in 64% of patients. Anxiety and depression were identified in 37% and 27% of patients. One-third of the patients were unable to ambulate independently. The median 6-minute walk distance was 56% of predicted. Only seven of the previously working patients (15%) had returned to work. Referral services and case management were provided 142 times. The median number of pharmacy interventions per patient was four.

B. Medication Management

1. Glucose dysregulation: A retrospective study of 74 patients with ARDS found that a blood glucose value of 153.5 mg/dL resulted in a 2.9 greater chance of developing cognitive impairment. A second retrospective case-control study of 37 surgical ICU patients with at least one episode of hypoglycemia found that cognitive impairment was higher in the hypoglycemic group ($p < 0.01$). Intensive insulin therapy (maintaining blood glucose levels between 80 and 100 mg/dL) in surgical ICU patients decreased neuropathy from 51.9% to 28.7%, and the prevalence of critical illness polyneuropathy (CIP) and critical illness myopathy (CIM) from 49% to 25% in surgical ICU patients ($p < 0.0001$) (Neurology 2005;64:1348-53). Intensive insulin therapy also decreased CIP and CIM from 51% to 39% in the medical ICU ($p = 0.02$) in patients who had an ICU stay of at least 1 week. The percentage of patients needing mechanical ventilation for at least 2 weeks was reduced from 42% to 32% in the surgical ICU ($p = 0.04$) and from 47% to 35% in the medical ICU ($p = 0.01$). Subsequently, NICE-SUGAR showed increased mortality in the intensive insulin group (81 to 108 mg/dL) (27.5%) versus conventional glucose control (less than 180 mg/dL) (24.5%) ($p = 0.02$). The SCCM guidelines for the use of an insulin infusion for the management of hyperglycemia in critically ill patients suggests that a blood glucose of 150 mg/dL or greater initiates interventions to maintain blood glucose less than 180 mg/dL and to avoid hypoglycemia based off the results of NICE-SUGAR.
2. Continuation of inappropriate medications: The frequency of prescribed potentially inappropriate medications (PIMs) and actually inappropriate medications (AIMs) was evaluated in a single-center study of 120 older adult ICU survivors. PIMs were defined as potentially harmful on the basis of prior studies and pharmacologic effects. PIMs could further be classified as AIMs if the benefit of the drug was considered less than the harm. The 2003 Beers Criteria and medication safety data published since 2003 were used to identify medications (Arch Intern Med. 2011;171:1032-4). Medications were identified at five points during the hospital stay: admission, ward admission, ICU admission, ICU discharge, and hospital discharge. The most common categories of PIMs identified at hospital discharge were opioids, anticholinergic medications, antidepressants, and drugs causing orthostasis. Thirty-six percent of these PIMs were considered AIMs. The PIM categories at hospital discharge with the highest positive predictive values for being AIMs were anticholinergics (55%), nonbenzodiazepine hypnotics (67%), benzodiazepines (67%), atypical antipsychotics (71%), and muscle relaxants (100%). In multivariate analysis, the number of discharge PIMs was independently predicted by the number of preadmission PIMs ($p < 0.001$), discharge to somewhere other than home ($p = 0.03$), and discharge from a surgical service ($p < 0.001$). Almost two-thirds of AIMs were initiated in the ICU.
3. Not restarting home medications: A large population-based Canadian cohort study of 396,380 patients evaluated records of hospital and outpatient medications prescribed from at least one of five of the following groups: (1) statins, (2) antiplatelet/anticoagulant agents, (3) levothyroxine, (4) respiratory inhalers, and (5) gastric acid-suppressing drugs. Patients were divided into three groups: hospitalization with an ICU admission, hospitalization without ICU admission, and nonhospitalized patients who served as the control group. Compared with control patients, those admitted to a hospital without an ICU stay were more likely to have medications discontinued among all five of the medication groups.

- Patients admitted to a hospital with an ICU stay were also more likely to have medications discontinued among all five of the medication groups than were control patients. Except for respiratory inhalers, there was a higher risk of medication discontinuation in all medication groups in patients hospitalized with an ICU admission than in patients hospitalized without an ICU admission. The composite outcome of death, hospitalization, and emergency department visit up to 1 year after hospital discharge in all study patients was higher in patients in whom a statin or antiplatelet or anticoagulant was discontinued (JAMA. 2011;306:840-7).
4. A prospective, observational cohort study was conducted of all outpatient appointments of a tertiary care hospital's post-ICU clinic between July 2012 and December 2015. The pharmacist completed medication reconciliation, interview, counseling, and resultant interventions during the post-ICU clinic appointment. The pharmacist did a full medication review in 56 of the patients (90%). All 56 patients had at least one pharmacy intervention. Medications were discontinued at the clinic appointment for 22 of the patients (39%). New medications were initiated in 18 of the patients (32%). An adverse drug event was identified in nine of the patients (18%). Adverse drug event preventive measures were implemented in 18 patients (32%). Thirteen patients (23%) had an influenza vaccination administered. Two patients (4%) received a pneumococcal vaccination (Ann Pharmacother 2018;52:713-23).
 5. A prospective, observational study was conducted in 12 ICUs/post-ICU clinics between September 2019 and July 2021. A full medication review was performed in 472 patients by a pharmacist at the post-ICU clinic. In 397 (84%) patients, pharmacy interventions were made. Medications were stopped in 124 (26%) and started in 91 (19%) patients. The number of patients that had a dose decreased was 51 (11%). The number of patients that had a dose increased was 43 (9%). Adverse drug event (ADE) preventive measures were initiated in 115 (24%) patients. Identification of ADEs occurred in 69 (15%) patients. In 30 (6%) patients, medication interactions were identified.

Patient Cases

5. T.L. (from question 4) was extubated 24 hours ago, is currently receiving dexmedetomidine 0.2 mcg/kg/hour, and has received two doses of fentanyl 25 mcg over 24 hours for pain. She is alert and calm with intermittent periods of agitation. Her pain score is now negative, and she is newly positive for delirium by CAM-ICU. Her laboratory values and vital signs are normal. Which would best be recommended for the management of delirium?
 - A. Continue dexmedetomidine, and start quetiapine for delirium.
 - B. Discontinue dexmedetomidine, and increase maintenance fluids for dehydration.
 - C. Discontinue dexmedetomidine, and order patient mobility as tolerated.
 - D. Continue dexmedetomidine, and schedule oxycodone sustained release every 12 hours.
6. P.V. is a 70-year-old woman intubated for severe respiratory failure (Fio₂ 80%) and refractory shock from methicillin-resistant *Staphylococcus aureus* pneumonia, for which she was administered antibiotics, vasopressors, and steroids. She is on day 5 of mechanical ventilation (Fio₂ 50%) and has been off vasopressors for 48 hours. The nurse describes PAD, but the patient denies pain. Medications include vancomycin 1000 mg daily, heparin 5000 units subcutaneously every 12 hours, hydrocortisone 50 mg every 6 hours, and fentanyl 75 mcg/hour. Which is the most appropriate recommendation at this time?
 - A. Increase fentanyl, and add midazolam for agitation.
 - B. Decrease fentanyl, and discontinue hydrocortisone.
 - C. Decrease fentanyl, and add haloperidol for delirium.
 - D. Increase fentanyl, and change vancomycin to linezolid.

VII. NEUROMUSCULAR BLOCKADE IN THE INTENSIVE CARE UNIT

- A. The most recent SCCM guidelines for the sustained use of neuromuscular blockade in the ICU were published in 2016. Surveys have reported a dramatic decrease in the use of NMBA during the past 20 years, from around 80% to 15% in patients on mechanical ventilation (Crit Care Med 2016;44:2079-103). This change in practice may be secondary to a better understanding of the serious adverse effects of prolonged paralysis, together with accepted standards of care for modes of mechanical ventilation in patients with ARDS.
- B. Clinical Scenarios for the Use of NMBA in the ICU May Include:
1. Rapid sequence intubation
 2. ARDS
 3. Status asthmaticus
 4. Elevated ICP
 5. Elevated intra-abdominal pressure
 6. Targeted temperature management after cardiac arrest
- C. Acute Respiratory Distress Syndrome
1. Cisatracurium has been the most-studied NMBA for ARDS since 2000, primarily as short-term treatment and in severe cases of ARDS. In 2010, a randomized placebo-controlled trial (n=340) found that short-term fixed-dose cisatracurium (48 hours) significantly improved 90-day survival, increased ventilator-free days, increased organ dysfunction-free days, and decreased barotrauma in patients with severe ARDS (P_{aO_2}/F_{iO_2} less than 120 mm Hg). The investigators found no difference in neuromuscular weakness compared with placebo. Other cisatracurium studies have shown improvements in oxygenation and a reduction in inflammatory mediators.
 2. In retrospective studies, the use of NMBA in ARDS was associated with a prolonged duration of mechanical ventilation, prolonged ICU length of stay, and increased mortality.
 3. Protective mechanisms of NMBA in severe ARDS: Researchers have proposed mechanisms by which an NMBA may protect the lung against further injury in severe ARDS. These mechanisms are not completely understood, but they may help explain the beneficial effects of NMBA in early, severe ARDS:
 - a. Provide improved adaptation to the ventilator through increased thoracopulmonary compliance.
 - b. Increase functional residual capacity, and decrease intrapulmonary shunt.
 - c. Provide uniform distribution of pulmonary perfusion and pressures, favoring the perfusion of ventilated areas.
 - d. Limit over-distention of high-compliance lung regions and recruit areas of smaller compliance.
 - e. Decrease muscular oxygen consumption by decreasing ventilator asynchrony.
 - f. Decrease production of proinflammatory cytokines in lungs and blood.
 - g. Provide protective role against ventilator-induced trauma, including decreased incidence of pneumothoraces.
 4. The 2016 SCCM NMBA guidelines suggest that a NMBA be administered by continuous intravenous infusion early in the course of ARDS for patients with a P_{aO_2}/F_{iO_2} of less than 150 mm Hg (weak recommendation). Use of NMBA in ARDS remains controversial. Short-term use of cisatracurium (48 hours or less) when used early may be beneficial for severe ARDS (P_{aO_2}/F_{iO_2} less than 120 mm Hg). The Reevaluation of Systemic Early Neuromuscular Blockade (ROSE) PETAL was a randomized, parallel-design study comparing cisatracurium with placebo for 48 hours on the effects of 90-day mortality. It was published after the guidelines and did not support using this early in course of ARDS.

D. Targeted Temperature Management After Cardiac Arrest

1. The 2016 SCCM NMBA guidelines make no recommendation on the routine use of NMBAs for patients undergoing therapeutic hypothermia after cardiac arrest (insufficient evidence). They recommend use of a protocol that includes guidance on NMBA administration in patients undergoing therapeutic hypothermia (good practice statement).
2. NMBAs have been used to prevent or treat shivering during therapeutic hypothermia.
3. The optimal combination and dosing of sedatives and paralytics have not been well established because the metabolism of these drugs is significantly slowed during hypothermia, and potency may be decreased. NMBAs have been used in both a bolus and a continuous infusion fashion during therapeutic hypothermia.

E. Sedation During NMBA: The 2016 SCCM NMBA guidelines recommend that optimal clinical practice require administering analgesic and sedative drugs before and during neuromuscular blockade with the goal of achieving deep sedation (good practice statement). They also suggest that patients receiving a continuous infusion of NMBA should receive a structured regimen of physiotherapy (weak recommendation). It is critical that patients be in a sedated, non-agitated, and pain-free state before initiating an NMBA. Once the patient becomes paralyzed from the NMBA, the ability to accurately assess mental status or pain is ostensibly challenging and often unattainable. The deeper the degree of paralysis, the higher the risk of drug accumulation because nurses cannot routinely complete sedation interruption or taper to a lighter level of sedation. Common scenarios that slow the clearance of sedatives (e.g., hepatic and renal failure or a hypothermic state) can add to the likelihood of increased drug exposure and delayed awakening times once the paralytic and sedatives are discontinued. This risk of drug accumulation underscores the importance of a daily assessment for need of paralysis and frequent tapering of NMBA dosing once it is safe for the patient.

F. Two Classes of NMBAs According to Mechanism of Action: Depolarizing and Nondepolarizing:

1. Depolarizing NMBAs: Bind and activate acetylcholine receptors, causing persistent depolarization, which then renders muscle fibers resistant to further cholinergic stimulation. Succinylcholine is the only available depolarizing NMBA. Because of its quick onset and short duration, it is commonly the drug of choice for urgent or emergency intubation.
 - a. Pharmacokinetics: Hydrolyzed by plasma pseudocholinesterase
 - b. Usual dose: 0.05–1.5 mg/kg intravenously or intramuscularly
 - c. Onset intravenously: 30–60 seconds; intramuscularly: 2–3 minutes
 - d. Duration intravenously: 4–6 minutes; intramuscularly: 10–30 minutes
 - e. Should not be used in patients with a history of malignant hyperthermia, hyperkalemia or risk factors for hyperkalemia (stroke, paralysis, or spinal, crush, or burn injuries after 24 hours), glaucoma, or penetrating eye injuries.
 - f. Adverse effects: Arrhythmias, bradycardia or tachycardia, hyperkalemia, rhabdomyolysis
2. Nondepolarizing NMBAs: Nicotinic receptor antagonists (competitive), blocking the action of acetylcholine at the neuromuscular junction. Divided into aminosteroid group (pancuronium, vecuronium, and rocuronium) and benzyl isoquinolinium group (atracurium, cisatracurium, doxacurium, and mivacurium).
 - a. Pancuronium: Long-acting aminosteroid; intermittent or scheduled bolus may be preferred to continuous infusion because of accumulation and variable clearance. Older NMBA, not used much in the United States.
 - i. Pharmacokinetics: Hepatically metabolized (30%–50%) and renally cleared as unchanged drug (50%–70%). Accumulation and prolonged duration of paralysis will occur with varying degrees of hepatic and/or renal dysfunction. Duration about 60–120 minutes.
 - ii. Adverse effects: Vagolytic activity, sympathetic stimulation, bradycardia, prolonged effect

- b. Vecuronium: Intermediate-acting aminosteroid; often used as a continuous infusion
 - i. Pharmacokinetics: Hepatically metabolized (30%–50%); cleared renally (20%–30%), with fecal excretion. Has an active metabolite, around half the activity of parent compound. Duration 30 minutes after bolus intubation dose.
 - ii. Adverse effects: Vagolytic activity at higher doses, prolonged weakness
 - iii. Can be reversed with sugammadex by forming a complex that decreases the amount of vecuronium available to bind to nicotinic receptors in the neuromuscular junction.
 - c. Rocuronium: An intermediate-acting aminosteroid; considered a suitable alternative to succinylcholine for rapid sequence intubation (dose: 0.6–1.2 mg/kg) because of its rapid onset of action (60–90 seconds). Duration 30–40 minutes.
 - i. Pharmacokinetics: Primarily hepatically metabolized, minimal renal excretion. No active metabolite. Prolonged effects have been observed in patients with hepatic or renal failure.
 - ii. Adverse effects: Vagolytic activity at higher doses, bradycardia
 - iii. Can be reversed with sugammadex by forming a complex that decreases the amount of rocuronium available to bind to nicotinic receptors in the neuromuscular junction.
 - d. Atracurium: Intermediate-acting benzyl isoquinolinium; a mixture of 10 stereoisomers (contains 15% cisatracurium)
 - i. Pharmacokinetics: Undergoes Hofmann elimination to form the toxic metabolite laudanosine at high levels. Laudanosine is a cerebral stimulant that may precipitate seizure activity, clearance dependent on liver and kidney function. Duration of atracurium 20–40 minutes.
 - ii. Adverse effects: Histamine release may cause cardiovascular adverse effects and bronchospasm; laudanosine accumulation may cause seizure activity.
 - e. Cisatracurium: An intermediate-acting benzyl isoquinolinium. Differences compared with atracurium: It is only one isomer, has a slower onset at normal bolus doses, no histamine release.
 - i. Pharmacokinetics: Undergoes Hofmann elimination, forms laudanosine but at much lower levels than atracurium. Renal and hepatic dysfunction do not alter cisatracurium clearance. Duration 30–60 minutes.
 - ii. Adverse effects: Prolonged weakness with continued use
- G. Drug Interactions with NMBAs: Certain medications may decrease the activity of NMBAs, whereas others can enhance or prolong the paralytic action.
- 1. Drugs decreasing the activity of NMBAs:
 - a. Calcium: Antagonizes the effect of magnesium on neuromuscular blockade
 - b. Carbamazepine: Competitor of acetylcholine receptor
 - c. Phenytoin: Depressed postsynaptic response to acetylcholine
 - d. Ranitidine: Unknown mechanism
 - e. Theophylline: Unknown mechanism
 - 2. Drugs prolonging the activity of NMBAs:
 - a. Antibiotics: Aminoglycosides, clindamycin, tetracyclines, vancomycin. Decreases presynaptic acetylcholine release with decreased postsynaptic acetylcholine receptor sensitivity; blocks acetylcholine receptor.
 - b. Cardiac medications: β -Blockers, calcium channel blockers, procainamide, quinidine, and furosemide. Decreases presynaptic acetylcholine release.
 - c. Immunosuppressants: Steroids (decrease end plate sensitivity to acetylcholine), cyclosporine (inhibits metabolism of certain NMBAs)

- H. Choice of NMBA: Intermediate- to longer-acting agents such as vecuronium may be tried in bolus fashion initially before continuous infusion, particularly if organ dysfunction is present. The duration of paralysis for NMBAs cleared by Hofmann degradation may be more reliable when used as a continuous infusion because their clearance is not dependent on renal or hepatic function.
- I. Train-of-Four (TOF) Monitoring and Dose Titration
1. The SCCM NMBA guidelines make no recommendation concerning the use of electroencephalogram-derived parameters (e.g., Bispectral Index [BIS], E-entropy, Cerebral State Index, and Patient State Index) as a measure of sedation during continuous NMBA administration (insufficient evidence). They suggest against the use of peripheral nerve stimulation (PNS) with TOF alone for monitoring the depth of neuromuscular blockade in patients receiving continuous NMBA infusions (very low quality evidence). They make no recommendation on the use of PNS to monitor the degree of block in patients undergoing therapeutic hypothermia (insufficient evidence). Typically, the goal of using an NMBA is to improve patient-ventilator synchrony and increase oxygenation. This may be achieved with varying degrees of paralysis and may not necessitate 100% block.
 2. Monitoring the depth of neuromuscular blockade by peripheral nerve stimulators (e.g., TOF), together with measured oxygenation parameters, helps find the “lowest effective paralytic dose” and allows quicker recovery of spontaneous neuromuscular transmission once the NMBA is discontinued. Some clinicians do not believe that TOF monitoring is necessary and believe that using the clinical values alone is sufficient to determine NMBA dosing.
 3. TOF delivers four supramaximal electrical impulses every 0.5 seconds to the ulnar, facial, or posterior tibial nerve. Response to the impulse is then measured by muscle twitches visualized from the associated innervated muscles (thumb or eye). Goals of paralysis can usually be reached with 2 or 3 of 4 twitches; 0 of 4 twitches indicates complete neuromuscular blockade, usually necessitating a decrease in NMBA dose. Oxygenation goals may be reached even with 4 of 4 twitches, indicating that the NMBA dose is effective and an increase is not warranted.
 4. A baseline electrical current should be established before initiating an NMBA to determine how much electrical current is needed to produce a twitch. Usually 10–20 mA (amperage) is sufficient. The conduction of the electrical impulse may be dampened because of peripheral edema, loss of electrode adhesion, incorrect electrode placement, and hypothermia, which can lead to inaccurate readings. These factors should be reassessed with each use of the TOF.
- J. Complications of NMBAs
1. Prolonged weakness: Several case reports associate the use of NMBAs and prolonged weakness, which could include myopathy, polyneuropathy, or neuromyopathy. Other risk factors may include concomitant use of corticosteroids, persistent hyperglycemia, and type of NMBA used. However, data are inconsistent and not controlled, and further studies are needed to clarify specific risk factors for prolonged weakness associated with NMBAs. Following a trend in creatine kinase concentration every 48–72 hours may help assess the presence of myopathy secondary to paralysis and prolonged immobilization. A creatine kinase concentration should not be solely relied on for the presence of myopathy, and daily determination of the need for the NMBA should still be considered, even with a normal creatine kinase.
 2. Corneal abrasions: Paralysis eliminates the ability of the eyes to close and blink, increasing the risk of corneal ulcerations and infection. Prophylactic eye protection must be used in all patients on NMBAs (e.g., lubricating eye ointments or eye covers).
 3. Thrombosis: Caused partly by immobility, patients receiving an NMBA may be up to 8 times more likely to have a DVT than those not on an NMBA. Prophylaxis for a DVT must be provided for all patients on an NMBA.

4. Awareness: Recent case reports document patient awareness during paralysis in the ICU. These patients report weird dreams, fear, resistance of restraints, thoughts of life and death, and pain. Patients must be deeply sedated before initiating an NMBA (Ann Emerg Med 2021;77:532-44).
5. Resistance to paralysis and/or potentiation: Certain disease states may produce an up-regulation in acetylcholine skeletal muscle receptors, leading to higher-than-normal doses of the NMBA (e.g., muscle trauma, muscle atrophy, burns). Acid-base disorders, electrolyte imbalances, and adrenal insufficiency may also cause unpredictable alterations in dosing requirements.
6. Anaphylaxis: Allergic reactions can occur after the first dose of an NMBA because the ammonium ions in NMBAs are commonly found in the household environment and in household products. If an allergic reaction is suspected, skin prick testing for the NMBA against a control can be done within 6 weeks of the reaction.

Patient Cases

7. A 55-year-old man intubated for severe ARDS (Pao₂/Fio₂ ratio less than 100 mm Hg) is receiving fentanyl 200 mcg/hour, midazolam 8 mg/hour, and propofol 40 mcg/kg/minute. He is deeply sedated but remains hypoxic and dyssynchronous with the ventilator after several changes in mechanical ventilation settings. Which is the most appropriate consideration at this time?
 - A. Start scheduled lorazepam every 6 hours.
 - B. Add quetiapine 50 mg every 8 hours.
 - C. Change propofol to dexmedetomidine.
 - D. Start a cisatracurium infusion.
8. A 70-year-old woman who is day 2 in the ICU is receiving a neuromuscular blocking agent (NMBA) and is sedated for severe ARDS. The TOF over 24 hours is 2 of 4 twitches at an amplitude of 10 mA. Arterial blood gas is pH 7.38, Pco₂ 40 mm Hg, Po₂ 91 mm Hg, and bicarbonate 24 mEq/L on 50% inspired oxygen and 10 cm H₂O PEEP; the patient is synchronous with the ventilator, and other clinical markers are stable. Which changes in management would be best to recommend?
 - A. Decrease stimulator amplitude to decrease pain from excessive electrical current.
 - B. Increase stimulator amplitude to test for more frequent twitches.
 - C. Decrease the NMBA dose because the patient is clinically stable.
 - D. Increase the NMBA dose until the TOF induces fewer twitches.

REFERENCES

Pain

1. Barr J, Fraser GL, Puntillo K, et al. Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med* 2013;41:263-306. The most updated guidelines from SCCM, the American College of Critical Care Medicine, and the American Society of Health-System Pharmacists. This was a multidisciplinary effort coordinating all three areas of management (pain, agitation, and delirium).
2. Breen D, Karabinis A, Malbrain M, et al. Decreased duration of mechanical ventilation when comparing analgesia-based sedation using remifentanyl with standard hypnotic-based sedation for up to 10 days in intensive care unit patients: a randomized trial. *Crit Care* 2005;9:R200-10.
3. Czarnecki M, Turner H, Collins PM, et al. Procedural pain management: a position statement with clinical practice recommendations. *Pain Manag Nurs* 2011;12:95-111.
4. Devlin JW, Skrobik Y, Gélinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med* 2018;46:e825-73.
5. Devabhakthuni S, Armahizer MJ, Dasta JF, et al. Analgosedation: a paradigm shift in intensive care unit sedation practice. *Ann Pharmacother* 2012;46:530-40. Review of literature using an analgesic-based method of sedation.
6. Duprey MS, Dijkstra-Kersten SM, Zaal IJ, et al. Opioid use increases the risk of delirium in critically ill adults independently of pain. *Am J Respir Crit Care Med* 2021;204:566-72.
7. Erstad BL, Puntillo K, Gilbert H, et al. Pain management principles in the critically ill. *Chest* 2009;135:1075-86. Comprehensive review of opiate use in the ICU; discusses a spectrum of issues, including acute versus chronic pain, adverse effects, non-opiate medications, nonpharmacologic methods of treating pain, and the withdrawal effects of opiates in the ICU.
8. Gelinas C, Fillion L, Puntillo K, et al. Validation of the critical-care pain observation tool in adult patients. *Am J Crit Care* 2006;15:420-7.
9. Kanji S, MacPhee H, Singh A, et al. Validation of the critical care pain observation tool in critically ill patients with delirium: a prospective cohort study. *Crit Care Med* 2016;44:943-7.
10. Liu J, Smith KE, Riker RR, et al. Methadone bioavailability and dose conversion implications with intravenous and enteral administration: a scoping review. *A J Health Syst Pharm* 2021;78:1395-401.
11. Mather LE. Clinical pharmacokinetics of fentanyl and its newer derivatives. *Clin Pharmacokinet* 1983;8:422-46.
12. McClave SA, Martindale RG, Vanek VW, et al. *JPEN J Parenter Enteral Nutr* 2009;33:277-316.
13. Mostafa SM, Bhandara S, Ritchie G, et al. Constipation and its implications in the critically ill patient. *Br J Anaesth* 2003;91:815-9.
14. Nguyen T, Frenette AJ, Johanson C, et al. Impaired gastrointestinal transit and its associated morbidity in the intensive care unit. *J Crit Care* 2013;28:537e11-7.
15. Park CM, Inouye SK, Marcantonio ER, et al. Perioperative gabapentin use and in-hospital adverse clinical events among older adults after major surgery. *JAMA Intern Med* 2022;182:1117-27.
16. Payen J, Bru O, Bosson J, et al. Assessing pain in critically ill sedated patients by using a behavioral pain scale. *Crit Care Med* 2001;29:2258-63. Validation study for the BPS.
17. Puntillo K, Ley SJ. Appropriately timed analgesics control pain due to chest tube removal. *Am J Crit Care* 2004;13:292-301.
18. Puntillo KA, Max A, Timsit J, et al. Determinants of procedural pain intensity in the intensive care unit. The Europain® study. *Am J Respir Crit Care Med* 2014;189:39-47.
19. Puntillo KA, Pasero C, Li D, et al. Evaluation of pain in ICU patients. *Chest* 2009;135:1069-74. Concise review on how to evaluate pain in the ICU patient who is nonverbal. Brief discussion on the validated pain scales for the adult ICU patient.
20. Puntillo KA, Wild LR, Morris AB, et al. Practices and predictors of analgesic interventions for adults undergoing painful procedures. *Am J Crit Care* 2002;11:415-29.
21. Rotondi AJ, Chelluri L, Sirio C, et al. Patients' recollections of stressful experiences while receiving prolonged mechanical ventilation in an

- intensive care unit. *Crit Care Med* 2002;30:746-52. Questionnaire study describing patients' recalled memories after being in the ICU and on mechanical ventilation for 48 hours or more.
22. Scholz J, Steinfaß M, Schulz M. Clinical pharmacokinetics of alfentanil, fentanyl, and sufentanil. An update. *Clin Pharmacokinet* 1996;31:275-92.
 23. Strøm T, Martinussen T, Toft P. A protocol of no sedation in critically ill patients receiving mechanical ventilation: a randomized controlled trial. *Lancet* 2010;375:475-80.
 24. Taylor D. Iatrogenic drug dependence—a problem in intensive care? *Intensive Crit Care Nurs* 1999;15:95-100. A good discussion on the potential for opiate and benzodiazepine dependence in patients who require long-term therapy of these agents in the ICU setting. Pharmacists can assist the ICU team in the appropriate dosing of opiates throughout the ICU stay and monitoring for symptoms of withdrawal while tapering off.
- Agitation**
1. Balas MC, Vaselevskis EE, Olsen KM, et al. Effectiveness and safety of the awakening and breathing coordination, delirium monitoring/management, and early exercise/mobility bundle. *Crit Care Med* 2014;42:1024-36. Known as the ABCDE trial, this was the second study to coordinate the efforts of SATs and SBTs. The investigators added delirium monitoring and management as well as an early mobility protocol to this study; medical or surgical ICU population, n=296. Primary outcome was median time breathing without mechanical ventilator assistance during the 28-day study period. This study used safety screens for the SAT, SBT, and mobility protocols. The intervention group had 3 more days of breathing without mechanical ventilator assistance than did the standard care group (median 24 vs. 21 days; p=0.04). The intervention group was almost half as likely to develop delirium (OR 0.55; p=0.03) and had increased odds of getting out of bed at least once during the ICU stay (p=0.003) compared with the standard care group.
 2. Barnes BJ, Gerst C, Smith JR, et al. Osmol gap as a surrogate marker for serum propylene glycol concentrations in patients receiving lorazepam for sedation. *Pharmacotherapy* 2006;26:23-33.
 3. Barr J, Fraser GL, Puntillo K, et al. Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med* 2013;41:263-306. The most updated guidelines from SCCM, the American College of Critical Care Medicine, and the American Society of Health-System Pharmacists.
 4. Beigmohammadi MT, Hanifeh M, Rouini MR, et al. Pharmacokinetic alterations in midazolam infusion versus bolus administration in mechanically ventilated critically ill patients. *Iran J Pharm Res* 2013;12:483-8.
 5. Bienvenu OJ, Gellar J, Althouse BM, et al. Post-traumatic stress disorder symptoms after acute lung injury: a 2-year prospective longitudinal study. *Psychol Med* 2013;43:2657-71.
 6. Brown C, Marotta J, Riker RR, et al. Prospective validation of sedation scale scores that identify light sedation: a pilot Study. *Am J Crit Care* 2022;31:202-8.
 7. Cammarono WB, Pittet JF, Weitz S, et al. Acute withdrawal syndrome related to the administration of analgesic and sedative medications in adult intensive care unit patients. *Crit Care Med* 1998;26:676-84.
 8. Carson SC, Kress JP, Rodgers J, et al. A randomized trial of intermittent lorazepam versus propofol with daily interruption in mechanically ventilated patients. *Crit Care Med* 2006;34:1326-32. Nice study from two large university hospitals, medical ICU population (n=546). Primary outcome was median days on mechanical ventilation, which was lower in the propofol daily interruption group (propofol = 5.8 days; intermittent lorazepam = 8.4 days; p=0.04). Hospital mortality was no different between the two groups.
 9. Cioccarl L, Luethia N, Bailey M, et al. The effect of dexmedetomidine on vasopressor requirements in patients with septic shock: a subgroup analysis of the Sedation Practice in Intensive Care evaluation [SPICE III] Trial *Crit Care* 2020;24:441.
 10. Devlin JW, Mallow-Corbett S, Riker RR. Adverse drug events associated with the use of analgesics, sedatives, and antipsychotics in the intensive care unit. *Crit Care Med* 2010;38:S231-43.
 11. de Wit M, Gennings C, Jenvey W, et al. Randomized trial comparing daily interruption of sedation and nursing-implemented sedation

- algorithm in medical intensive care unit patients. *Crit Care* 2008;12:1-9. Pivotal study terminated early because of complications in patients randomized to the daily interruption of sedation group. The authors concluded that daily interruption is not appropriate in all medical ICU patients. A patient safety screen is now recommended for all patients as part of any daily interruption or SAT.
12. Fodale V, La Monaca E. Propofol infusion syndrome: an overview of a perplexing disease. *Drug Saf* 2008;31:293-303.
 13. Fraser GL, Devlin J, Worby CP, et al. Benzodiazepine versus nonbenzodiazepine-based sedation for mechanically ventilated critically ill adults: a systematic review and meta-analysis of randomized trials. *Crit Care Med* 2013;41:S30-8. Good review of trials comparing the major outcomes of drugs used for sedation: days on mechanical ventilation, days in the ICU, incidence of delirium, and mortality.
 14. Galvin M, Jago-Byrne MC, Fitzsimons M, et al. *Int J Clin Pharm* 2013;35:14-211. Clinical pharmacist's contribution to medication reconciliation on admission to a hospital in Ireland.
 15. Girard TD, Kress JP, Fuchs BD, et al. Efficacy and safety of a paired sedation and ventilator weaning protocol for mechanically ventilated patients in intensive care (Awakening and Breathing Controlled trial): a randomized, controlled trial. *Lancet* 2008;371:126-34. Known as the Wake Up and Breathe, or ABC, trial, this was the first study to coordinate SATs with SBTs. The study was from four tertiary care hospitals (n=336), and primary outcome was time breathing without assistance during a 28-day study period. This study used a safety screen for both the SAT and the SBT protocols. The intervention group spent more time breathing without assistance than did the standard care group (median 14.7 vs. 11.6 days; p=0.02).
 16. Grayson KE, Bailey M, Balachandran M, et al. The effect of early sedation with dexmedetomidine on body temperature in critically ill patients. *Crit Care Med* 2021;49:1118-28.
 17. Herridge MS, Tansey CM, Matte A, et al. Functional disability 5 years after acute respiratory distress syndrome. *N Engl J Med* 2011;364:1294-304.
 18. Honiden S, Siegel M. Analytic reviews: managing the agitated patient in the ICU: sedation, analgesia, and neuromuscular blockade. *J Intensive Care Med* 2010;25:187-204. Succinct review of the primary issues in sedation management. This was published before the 2013 PAD guidelines, but it contains a very good discussion.
 19. Horinek EL, Kiser TH, Fish DN, et al. Propylene glycol accumulation in critically ill patients receiving continuous infusion lorazepam infusions. *Ann Pharmacother* 2009;43:1964-71.
 20. Hughes CG, Girard TD, Pandharipande P. Daily sedation interruption versus targeted light sedation strategies in ICU patients. *Crit Care Med* 2013;41:S39-45. This review finds that it is still unclear whether one sedation strategy is better than the other, that using the two strategies together may offer more benefit than either alone, and that coordinating either strategy with SBT should be considered.
 21. Jakob SM, Ruokonen E, Grounds RM, et al. Dexmedetomidine versus midazolam or propofol for sedation during prolonged mechanical ventilation. *JAMA* 2012;307:1151-60. This article reported on two trials; the "MIDEX" trial compared midazolam with dexmedetomidine, and the "PRODEX" trial compared propofol with dexmedetomidine. Only fentanyl boluses were used for pain management. Primary outcomes were time at target sedation level and duration of mechanical ventilation. The time at target sedation level was the same between groups from both studies; the only difference in median time on mechanical ventilation was between midazolam and dexmedetomidine (164 hours vs. 123 hours; p=0.03).
 22. Kane-Gill SL, Jacobi J, Rothschild JM. Adverse drug events in the intensive care units: risk factors, impact, and the role of team care. *Crit Care Med* 2010;38:S83-89.
 23. Kawazoe Y, Miyamoto K, Morimoto T, et al. Effect of dexmedetomidine on mortality and ventilator-free days in patients requiring mechanical ventilation with sepsis: a randomized clinical trial. *JAMA* 2017;317:1321-8.
 24. Kukoyi AT, Coker SA, Lewis LD. Two cases of acute dexmedetomidine withdrawal syndrome following prolonged infusion in the intensive care unit: report of cases and review of the literature. *Hum Exp Toxicol* 2013;32:106-110.
 25. Lewis K, Alshamsi F, Carayannopoulos KL, et al. Dexmedetomidine vs other sedatives in critically ill mechanically ventilated adults: a systematic

- review and meta-analysis of randomized trials. *Intensive Care Med* 2022;48:811-40.
26. Mehta S, Burry L, Cook D, et al. Daily sedation interruption in mechanically ventilated critically ill patients cared for with a sedation protocol. *JAMA* 2012;308:1985-92. This study (for the SLEAP study investigators and Canadian critical care trials group) compared daily sedation interruption with a standard sedation protocol targeting lighter sedation scores (e.g., RASS 0 to -3; SAS 3 or 4). The study found no difference between the two groups in their end points of time on mechanical ventilation or duration of ICU stay.
27. Molina-Infante J, Arias A, Vara-Brenes D, et al. Propofol administration is safe in adult eosinophilic esophagitis patients sensitized to egg, soy, or peanut. *Allergy* 2014;69:388-94.
28. Moller MH, Alhazzani W, Lewis K, et al. Use of dexmedetomidine for sedation in mechanically ventilated adult ICU patients: a rapid practice guideline. *Intensive Care Med* 2022;48:801-10.
29. Pandharipande PP, Pun BT, Herr DL, et al. Effects of sedation with dexmedetomidine versus lorazepam on acute brain dysfunction in mechanically ventilated patients: the MENDS randomized controlled trial. *JAMA* 2007;298:2644-53.
30. Pandharipande PP, Shintani A, Peterson J, et al. Lorazepam is an independent risk factor for transitioning to delirium in intensive care unit patients. *Anesthesiology* 2006;104:21-6. One of the first studies specifically addressing the use of lorazepam as a risk of developing delirium.
31. Riker R, Shahabi Y, Bokesch P, et al. Dexmedetomidine versus midazolam for sedation of critically ill patients. A randomized trial: SEDCOM. *JAMA* 2009;301:489-99. This study investigated differences in time at targeted sedation level between dexmedetomidine and midazolam. The investigators found no difference in efficacy of sedation; the prevalence of delirium was lower in the dexmedetomidine group, and the median time to extubation was shorter in the dexmedetomidine group.
32. Seymour CW, Pandharipande PP, Koestner T, et al. Diurnal sedative changes during intensive care: impact on liberation from mechanical ventilation and delirium. *Crit Care Med* 2012 40:2788-96.
33. Shafer A. Complications of sedation with midazolam in the intensive care unit and a comparison with other sedative regimens. *Crit Care Med* 1998;26:947-56.
34. Shehabi Y, Bellomo R, Reade MC et al. Early intensive care sedation predicts long-term mortality in ventilated critically ill patients. *Am J Respir Crit Care Med*. 2012;15:724-31.
35. Shehabi Y, Howe BD, Bellomo R, et al. Early sedation with dexmedetomidine in critically ill patients. *N Engl J Med* 2019;380:2506-17.
36. Swart EL, de Jongh J, Zuideveld KP, et al. Population pharmacokinetics of lorazepam and midazolam and their metabolites in intensive care patients on continuous venovenous hemofiltration. *Am J Kidney Dis* 2005;45:360-71.
37. Uijtendaal EV, van Harssel LL, Hugenholtz GW, et al. Analysis of potential drug-drug interactions in medical intensive care unit patients. *Pharmacotherapy* 2014;34:213-9.
38. Verbeek RK. Pharmacokinetics and dosage adjustment in patients with hepatic dysfunction. *Eur J Clin Pharmacol* 2008;64:1147-61.
39. Vonbach P, Dubied A, Krähenbühl S, et al. Prevalence of drug-drug interactions at hospital entry and during hospital stay of patients in internal medicine. *Eur J Intern Med* 2008;19:413-20.

Delirium

1. Adler J, Malone D. Early mobilization in the intensive care unit: a systematic review. *Cardiopulm Phys Ther J* 2012;23:5-13. Good review of current studies that investigated the effects of early mobilization of patients in the ICU on functional outcomes. More research is ongoing in this area, particularly as early mobility pertains to cognitive function and delirium.
2. Andersen-Ranberg NC, Poulsen LM, Perner A, et al. Haloperidol for the treatment of delirium in ICU patients. *N Engl J Med* 2022;387:2425-35.
3. Brummel NE, Jackson JC, Pandharipande PP, et al. Delirium in the intensive care unit and subsequent long-term disability among survivors of mechanical ventilation. *Crit Care Med* 2014;42:369-77.
4. Brummel NE, Vasilevskis EE, Han JH, et al. Implementing delirium screening in the ICU: secrets to success. *Crit Care Med* 2013;41:2196-208.

5. Devlin JW, Roberts RJ, Fong JJ, et al. Efficacy and safety of quetiapine in critically ill patients with delirium: a prospective, multicenter, randomized, double-blind, placebo-controlled pilot study. *Crit Care Med* 2010;38:419-27.
6. Devlin JW, Zaal I, Slooter A. Clarifying the confusion surrounding drug-associated delirium in the ICU. *Crit Care Med* 2014;42:1565-66. Discusses issues with current delirium research and latest findings from the corticosteroid-associated delirium study by Schreiber MP, et al. (*Crit Care Med* 2014;42:1480-6).
7. Girard TD, Exline MC, Carson SS, et al. Haloperidol and ziprasidone for treatment of delirium in critical illness. *N Engl J Med* 2018; 379:2506-16.
8. Girard TM, Pandharipande P, Carson SS, et al. Feasibility, efficiency, and safety of antipsychotics for intensive care unit delirium: the MIND randomized, placebo-controlled trial. *Crit Care Med* 2010;38:428-37. Pilot trial showing that treatment with antipsychotics in a 21-day study did not improve the number of days alive without delirium or coma.
9. Girard TM, Pandharipande P, Ely W. Delirium in the intensive care unit. *Crit Care* 2008;12(suppl 3):1-9. Good review of pathophysiology and risk factors for delirium. Also has a discussion on how to monitor, prevent, and treat delirium.
10. Jackson P, Khan A. Delirium in critically ill patients. *Crit Care Clin* 2015;31:589-603.
11. Jeon K, Jeong BH, Ko MG, et al. Impact of delirium on weaning from mechanical ventilation in medical patients. *Respirology* 2016;21:313-20.
12. Kamdar BB, King LM, Collop NA, et al. The effect of a quality improvement intervention on perceived sleep quality and cognition in a medical ICU. *Crit Care Med* 2013;41:800-9. This study investigated the effects of a sleep protocol before and after implementation. Study investigators found that implementation of a sleep protocol is feasible in an adult ICU, decreases perceived level of noise, and may decrease the incidence of delirium. Very good study that moves forward the discussion regarding the importance of adequate sleep in the ICU and for future research in this area.
13. Kees J, Kalisvaart MD, Jos FM, et al. Haloperidol prophylaxis for elderly hip surgery patients at risk for delirium: a randomized placebo controlled study. *J Am Geriatr Soc* 2005;53:1658-66.
14. Kram BR, Kram SJ, Brooks KR. Implications of atypical antipsychotic prescribing in the intensive care unit. *J Crit Care* 2015;30:814-8.
15. Lichtenbelt BJ, Olofson E, Van Kleef JW, et al. Propofol reduces the distribution and clearance of midazolam. *Anesth Analg* 2010;110:1597-606.
16. Marshall J, Herzig SJ, Howell MD, et al. Antipsychotic utilization in the intensive care unit and in transitions of care. *J Crit Care* 2016;33:119-24.
17. Mehta S, Cook D, Devlin JW, et al. Prevalence, risk factors, and outcomes of delirium in mechanically ventilated adults. *Crit Care Med* 2015;43:557-66.
18. Pandharipande PP, Girard TD, Jackson JC, et al. Long-term cognitive impairment after critical illness. *N Engl J Med* 2013;369:1306-16. This trial is from the BRAIN-ICU investigators group, which studied the risk factors for and incidence of cognitive impairment 3 and 12 months from ICU discharge from a medical or surgical ICU. They found that 74% of patients developed delirium during their ICU stay, and a longer duration of delirium was associated with worse global cognition and executive function scores at 3 and 12 months after discharge. Use of sedative and analgesic medications was not associated with cognitive impairment in this study at 3 and 12 months.
19. Patel SB, Poston JT, Pohlman A, et al. Rapidly reversible, sedation-related delirium versus persistent delirium in the intensive care unit. *Am J Respir Crit Care Med* 2014;189:658-65.
20. Pisani MA, Friese RS, Gehlbach BK, et al. Sleep in the intensive care unit. *Am J Respir Crit Care Med* 2015;191:731-8.
21. Pun BT, Balas MC, Barnes-Daly MA, et al. Caring for critically ill patients with the ABCDEF bundle: results of the ICU liberation collaborative in over 15,000 adults. *Crit Care Med* 2018 Oct 18. [Epub ahead of print]
22. Schreiber MP, Colantuoni E, Bienvenu O, et al. Corticosteroids and transition to delirium in patients with acute lung injury. *Crit Care Med* 2014;42:1480-6.
23. Skrobik Y, Duprey MS, Hill NS, et al. Low-dose nocturnal dexmedetomidine prevents ICU delirium: a randomized, placebo-controlled trial. *Am J Respir Crit Care Med* 2018;197:1147-56.

24. Sosnowski K, Lin F, Mitchell ML. Early rehabilitation in the intensive care unit: an integrated literature review. *Aust Crit Care* 2015;28:216-25.
 25. Stollings JL, Boneyk CB, Birdrow CI, et al. antipsychotics and the QTc interval during delirium in the intensive care unit a secondary analysis of a randomized clinical trial. *JAMA Netw Open*. 2024;7:e2352034.
 26. Van den Boogaard M, Slooter AJC, Bruggemann FJM, et al. Effect of haloperidol on survival among critically ill adults with a high risk of delirium: the REDUCE randomized clinical trial. *JAMA* 2018;319:680-90.
 27. Vanderbilt University Medical Center. ABCDEs of Prevention and Safety. Available at www.icudelirium.org. Vanderbilt website containing extensive educational materials about delirium in the ICU for medical professionals, patients, and families. This website uses several modalities for education, including references, training manuals, Vanderbilt's protocols, and videos, to assist medical professionals in assessing and managing delirium in their ICU.
 28. West S, Kenedi C. Strategies to prevent the neuropsychiatric side effects of corticosteroids: a case report and review of the literature. *Curr Opin Organ Transplant* 2014;19:201-8.
 29. Wilcox ME, Brummel NE, Archer K, et al. Cognitive dysfunction on ICU patients: risk factors, predictors, and rehabilitation interventions. *Crit Care Med* 2013;41:S81-S98.
 30. Wolters AE, Veldhuijzen DS, Peelen LM, et al. Systemic corticosteroids and transition to delirium in critically ill patients. *Crit Care Med* 2015;43:e585-8.
 31. Wolters AW, Zaal IJ, Veldhuijzen DS, et al. Anticholinergic medication use and transition to delirium in critically ill patients: a prospective cohort study. *Crit Care Med* 2015;42:1846-52.
 32. Zaal IJ, Devlin JW, Haelbag M, et al. Benzodiazepine-associated delirium in critically ill adults. *Intensive Care Med* 2015;41:2120-7.
 33. Zaal IJ, Devlin JW, Peelen LM, et al. A systematic review of risk factors for delirium in the ICU. *Crit Care Med* 2015;43:40-7. Most updated review of risk factors for delirium in adult ICU patients.
- ### Post Intensive Care Syndrome
1. Duning T, van den Heuvel I, Dickmann A, et al. Hypoglycemia aggravates critical illness-induced neurocognitive dysfunction. *Diabetes Care* 2010;33:639-44
 2. Hermans G, Wilmer A, Meersseman W, et al. Impact of intensive insulin therapy on neuromuscular complications and ventilator dependency in the medical intensive care unit. *Am J Respir Crit Care Med* 2007;175:480-9
 3. Herridge MS, Cheung AM, Tansey CM, et al. One-year outcomes in survivors of the acute respiratory distress syndrome. *N Engl J Med* 2003;348:683-93
 4. Herridge MS, Tansey CM, Matte A, et al. Functional disability 5 years after acute respiratory distress syndrome. *N Engl J Med* 2011;364:1293-304
 5. Hopkins RO, Suchyta MR, Snow GL, et al. Blood glucose dysregulation and cognitive outcome in ARDS survivors. *Brain Inj* 2010;24:1478-84
 6. Jackson JC, Pandharipande PP, Girard TD, et al. Depression, post-traumatic stress disorder, and functional disability in survivors of critical illness in the BRAIN-ICU study: a longitudinal cohort study. *Lancet Respir Med*. 2014 May;2(5):369-79.
 7. Sevin CM, Bloom SL, Jackson JC, et al. Comprehensive care of ICU survivors: development and implementation of an ICU recovery center. *J Crit Care* 2018;46:141-8.
 8. Stollings JL, Bloom SL, Huggins SL, et al. Medication management to ameliorate post intensive care syndrome. *AACN Adv Crit Care* 2016;27:133-40.
 9. Stollings JL, Bloom SL, Wang L, et al. Critical care pharmacists and medication management in an ICU recovery center. *Ann Pharmacother* 2018;52:713-23.
 10. Stollings JL, Caylor MM. Post intensive care syndrome and the role of a follow-up clinic. *Am J Health Syst Pharm* 2015;72:1315-23.
 11. Stollings JL, Poyant JO, Groth CM, et al. An international, multicenter evaluation of comprehensive medication management by pharmacists in ICU recovery centers. *J Intensive Care Med* 2023;38:957-65.
 12. Van den BG, Schoonheydt K, Becx P, et al. Insulin therapy protects the central and peripheral nervous system of intensive care patients. *Neurology* 2005;64:1348-53

Neuromuscular Blocking Agents

1. Greenburg SB, Vender J. The use of neuromuscular blocking agents in the ICU: where are we now? *Crit Care Med* 2013;41:1332-44. Very good review and discussion of NMDAs, including indications, pharmacology, TOF monitoring, and complications of prolonged paralysis.
2. Hraiech S, Dizier S, Papazian L. The use of paralytics in patients with acute respiratory distress syndrome. *Clin Chest Med* 2014;35:753-63.
3. Huang DT, Angus DC, Moss M, et al. Design and rationale for the reevaluation of systemic early neuromuscular blockade trial for acute respiratory distress syndrome. *Ann Am Thorac Soc* 2017;14:124-33.
4. Lagneau F, D'honneur G, Plaud B, et al. A comparison of two depths of prolonged neuromuscular blockade induced by cisatracurium in mechanically ventilated critically ill patients. *Intensive Care Med* 2002;28:1735-41.
5. Murray MJ, DeBlock H, Erstad B, et al. Clinical practice guidelines of the sustained neuromuscular blockade in the adult critically ill patient. *Crit Care Med* 2016;44:2079-103.
6. National Heart, Lung, and Blood Institute PETAL Clinical Trials Network. Early neuromuscular blockade in the acute respiratory distress syndrome. *N Engl J Med* 2019;380:1997-2008.
7. Papazian L, Forel JM, Gacouin A, et al. Neuromuscular blockers in early acute respiratory distress syndrome. *N Engl J Med* 2010;363:1107-16. Large multicenter, double-blind trial completed in France (n=340); studied the effects of cisatracurium in patients with early and severe ARDS. Primary end point was 90-day mortality. Investigators found that a fixed dose of cisatracurium for 48 hours decreased mortality in patients with severe ARDS (P_{aO_2}/F_{iO_2} less than 120 mm Hg).
8. Tortorici MA, Kochanek PM, Poloyac SM. Effects of hypothermia on drug disposition, metabolism, and response: a focus of hypothermia-mediated alterations on the cytochrome P450 enzyme system. *Crit Care Med* 2007;35:2196-204.

ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: D**

This patient has a clear indication for intravenous pain medication from his recent trauma and multiple fractures. An “as-needed” opiate would likely not keep up with his pain control needs (Answers B and C are incorrect). His age and history of hypertension place him at risk of delirium; therefore, a benzodiazepine is not the best initial choice for this patient. A fentanyl infusion for pain with a propofol infusion if needed for sedation is the most appropriate answer (Answer A is incorrect; Answer D is correct).

2. **Answer: D**

Chest tube removal is specifically cited in the PADIS guidelines as an indication for both preemptive analgesia and nonpharmacologic relaxation techniques. This is given a “strong” recommendation, determining that the benefits outweigh the risks of preemptive therapy (Answer D is correct). Acetaminophen given just before the procedure will most likely not adequately treat pain associated with chest tube removal (Answer A is incorrect). Increasing an opiate infusion several hours before a bedside procedure can expose the patient to substantially higher amounts of drug than needed and cause delayed awakening times or other significant adverse effects from opiates (Answer C is incorrect). Extensive studies of appropriate preemptive analgesia for chest tube removal have not been completed; however, administering an opiate appropriately timed before manipulation of a chest tube is an accepted standard of therapy (Answer B is incorrect).

3. **Answer: B**

This patient has end-stage liver failure and acute renal failure; he will therefore not predictably clear midazolam or fentanyl infusions. With a RASS of -4 to -5 , indicating no meaningful responsiveness to stimuli, all sedatives should be held if the patient is otherwise clinically stable to allow time for clearance of medications. If the patient was to develop pain based on a CPOT of 3 or more, then intermittent fentanyl is appropriate (Answer B is correct). A decrease in sedative dose or changing to a different sedative is not needed at this time based on the deeply sedated RASS score and would only further delay awakening time. Intermittent doses of benzodiazepines should be avoided. (Answers A, C, and D are incorrect).

4. **Answer: B**

Withdrawal from certain home medications may occur if these medications are not reinitiated within a few days of admission. The onset of withdrawal symptoms will vary depending on the half-life of each medication. Symptoms may include agitation, anxiety, psychosis, insomnia, hypertension, and tachycardia and can occur with medications such as opiates, GABA receptor agonists, antiepileptics, antidepressants, and antipsychotics. A pharmacist can assist the medical team by obtaining a thorough medication history and assessment of home medication adherence to help identify drug withdrawal symptoms. Reinitiating these medications can be considered, unless contraindicated because of the clinical scenario (e.g., drug-drug interactions, drug-disease state interactions). The Agitation and Sedation section of the PADIS guidelines discusses identifying and treating the etiology of agitation before adding other medications; reinitiating the benzodiazepine and antidepressant to treat withdrawal symptoms is the most appropriate answer (Answer B is correct). Neither fentanyl nor dexmedetomidine would treat withdrawal from a benzodiazepine or antidepressant (Answers A, C, and D are incorrect).

5. **Answer: C**

The PADIS guidelines stress using nonpharmacologic means to manage delirium when it is safe for the patient. Strong evidence for using dexmedetomidine to treat delirium is still not available and the MIND USA Study has shown that anti-psychotics are ineffective at treating delirium. This patient’s presentation of “alert and calm with intermittent periods of agitation” is a common scenario, and initial therapy should focus on reorienting and getting the patient interactive and mobile (Answers A and D are incorrect; Answer C is correct). Dehydration is a common cause of agitation, and it should be addressed; however, with normal laboratory values and vital signs, this patient is unlikely dehydrated at this time (Answer B is incorrect).

6. **Answer: B**

In the general population, systemic corticosteroids are known to cause many neuropsychiatric events, including hyperactivity and agitation; in a recent study of adult ICU patients with acute lung injury, only age and use of systemic corticosteroids in the preceding 24 hours were

independently associated with the transition to delirium from a non-delirious state (Answer B is correct). Benzodiazepines have a sedating effect and may calm an acutely agitated patient; however, they would not be recommended in this patient because they could worsen her confusion or delirium (Answer A is incorrect). The PADIS guidelines state that no evidence supports the use of haloperidol to reduce the duration of delirium (Answer C is incorrect). Vancomycin is not currently recognized as a cause of delirium; therefore, changing to linezolid is not indicated (Answer D is incorrect).

7. Answer: D

The midazolam dose is high, and the patient is “deeply sedated”; therefore, adding another benzodiazepine will likely not improve this patient’s clinical status. Quetiapine has no indication for general sedation in a critically ill patient, and it should not be a consideration for sedation in this patient with severe ARDS. Dexmedetomidine is considered a weak sedative with no effect on respiratory drive; therefore, it would likely not improve this patient’s ventilator dyssynchrony and hypoxia (Answers A–C are incorrect). At this stage in the patient’s clinical course, it is reasonable to consider an NMBA. In a 2010 study of cisatracurium versus placebo for 48 hours in early ARDS, the cisatracurium group had more days free of mechanical ventilation and decreased mortality (30% vs. 44%) at 90 days for the subgroup of patients with severe ARDS ($\text{Pao}_2/\text{Fio}_2$ ratio less than 120 mm Hg). The incidence of pneumothorax was lower in the cisatracurium group than in the placebo group (4% vs. 11%). There were more days free of organ failure (non-lung) in the cisatracurium group than in the placebo group (15.8 vs. 12.2 days) in the first 28 days. A recent meta-analysis concluded that using short-term cisatracurium in patients with severe ARDS decreases mortality and time on mechanical ventilation compared with placebo. The risk of prolonged neuromuscular weakness was not found in these studies; however, use beyond 48 hours may increase this risk (Answer D is correct).

8. Answer: C

The TOF method of assessment is primarily used to help determine the degree of neuromuscular blockade and should not be used to titrate the dose of the NMBA. The patient’s clinical status and laboratory values are the true determinants for dose adjustment of the NMBA. Patients may be at their clinical goal with a TOF of 2 or 3 twitches of 4. This is the ideal scenario, and it will

predict a faster reversal of neuromuscular blockade (Answer C is correct). A TOF of 0 or 1 of 4 twitches predicts a significantly slower neuromuscular recovery time, and clinicians should try to decrease the NMBA as soon as the patient is clinically stable by laboratory values and ventilator management (Answer D is incorrect). A baseline electrical current intensity (amperage) should be established before the onset of neuromuscular blockade and should not be changed during paralysis unless a new baseline is indicated (Answer A is incorrect). As the electrical intensity (amperage) is established, an increase in the amperage is not indicated during infusion of the NMBA in order to increase the number of twitches. A decrease in the dose of NMBA would be indicated if an increase in the number of twitches were the clinical goal (Answer B is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: D**

Propofol infusion syndrome is a well-documented and complex set of adverse events, potentially resulting in multiorgan failure. An elevation in lactate, creatine kinase, transaminases, SCr, and triglycerides and the presence of a metabolic acidosis are some of the abnormalities that should concern the critical care provider for the presence of PRIS (Answer D is correct). Both DVT and critical illness polyneuropathy are serious concerns in the ICU patient; however, the abnormalities in the case are not representative of these complications (Answers A and B are incorrect). There are currently no known abnormal laboratory values to help determine whether delirium is present in an ICU patient (Answer C is incorrect).

2. **Answer: B**

This patient is at risk of propylene glycol toxicity after receiving a lorazepam drip for more than 48 hours. Lorazepam is dissolved in propylene glycol, an alcohol that can induce an osmolar gap and metabolic lactic acidosis, particularly in patients with significant hepatic or renal failure. Quantitative propylene glycol levels may be unavailable; therefore, surrogate markers such as an abnormal osmolar gap (greater than 10 mmol) and metabolic acidosis may indicate propylene glycol toxicity and a need to discontinue lorazepam. Although lorazepam drips are not routinely used for general sedation in adult ICUs, they may be used for other indications (e.g., severe EtOH [ethyl alcohol] or benzodiazepine withdrawal), and clinicians should remain aware of this serious complication (Answer B is correct). With an oxygen saturation of 98% on 2 L of oxygen, this patient does not meet the predefined criteria of ARDS (Answer A is incorrect). Encephalopathy will not cause a metabolic acidosis; therefore, an ammonia concentration would not be helpful at this stage (Answer C is incorrect). With a low fractional excretion of sodium and a high BUN/SCr ratio, the patient's laboratory values are indicative of a pre-renal concern versus acute tubular necrosis (Answer D is incorrect).

3. **Answer: D**

It is inappropriate to initiate an NMBA in a patient who has a sedation score indicating "agitation." This implies that the patient may potentially detect pain or discomfort

while paralyzed (Answers B and C are incorrect). The goal should be to achieve a deeply sedated and/or non-agitated state before initiating an NMBA in an effort to avoid any patient discomfort that may be undetected during paralysis (Answer D is correct). The SAT would be inappropriate in someone who is rated "agitated" on the sedation scale or in a patient requiring escalating doses of sedation (Answer A is incorrect).

4. **Answer: B**

The pharmacokinetics/dynamics of prolonged fentanyl infusions have not been well described in the adult ICU population. Most data for fentanyl are derived from short-term infusions or boluses in healthy volunteers and in animal models. Fentanyl is hepatically metabolized primarily by the CYP3A4 enzyme, and decreased clearance of fentanyl has been described in patients with significant liver disease. Other properties of fentanyl (e.g., high volume of distribution, high protein binding, and high lipophilicity) may contribute to unpredictable clearance and a prolonged context-sensitive half-time for patients in acute renal failure or in patients who have inadequate nutritional status (Answer B is correct). Propofol is a CYP3A4 inhibitor; therefore, it should not induce the metabolism of fentanyl (Answer C is incorrect). Propofol is known to chelate trace elements and increase urinary loss of zinc when used for more than 5 days; propofol has not been shown to cause hypocalcemia (Answer D is incorrect). Disease states identified as risk factors for PRIS may include sepsis, acute liver failure, and history of pancreatitis; ARDS is not currently a documented risk factor (Answer A is incorrect).

5. **Answer: B**

Withdrawal from certain home medications may occur if these medications are not reinitiated within a few days of admission. The onset of withdrawal symptoms will vary depending on the half-life of each medication. Symptoms may include agitation, anxiety, psychosis, insomnia, hypertension, and tachycardia and can occur with medications such as opiates, GABA receptor agonists, antiepileptics, antidepressants, and antipsychotics. A pharmacist can assist the medical team by obtaining a thorough medication history and assessment of home medication compliance to help identify drug withdrawal symptoms. Reinitiating these medications

can be considered unless contraindicated because of the clinical scenario (e.g., drug-drug interactions, drug-disease state interactions). The Agitation and Sedation section of the PADIS guidelines discusses identifying and treating the etiology of agitation before adding other medications; fentanyl should treat this patient's chronic pain and treat opiate/tramadol withdrawal (Answer B is correct). Adding quetiapine or lorazepam for agitation would not address or treat the underlying etiology of potential opiate/tramadol withdrawal (Answers A and C are incorrect). Patient-controlled analgesia with an opiate would help treat opiate withdrawal; however, this patient is not alert enough to use it (Answer D is incorrect).

6. Answer: D

Up to 30%–60% of ICU patients may go through alcohol withdrawal on cessation of alcohol use. The presence of alcohol withdrawal in ICU patients may prolong their ICU stay, increase hospital costs significantly, and lead to other complications during the hospital stay. Early and aggressive symptom-triggered management with a benzodiazepine, particularly in a patient with a history of alcohol withdrawal, is a key element in management. There is no otherwise published protocol for alcohol withdrawal in ICU patients. In this case, although dexmedetomidine is useful as an adjunctive agent to help decrease sympathetic storm and agitation secondary to alcohol withdrawal, it is not currently recommended for use as a single agent for alcohol withdrawal (Answer A is incorrect). Opiates do not treat alcohol withdrawal (Answer C is incorrect). Phenytoin is a known anti-epileptic for use in epilepsy; however, it has not been shown to be effective in preventing or treating alcohol withdrawal (Answer B is incorrect). Benzodiazepines are the drugs of choice for alcohol withdrawal seizures; therefore, midazolam is the most appropriate drug for this patient, whose medical history is significant for recurrent alcohol withdrawal seizures (Answer D is correct).

7. Answer: B

This patient is at risk of delirium given the patient's age and being critically ill in the ICU. Avoiding medications that may cause or worsen delirium, such as benzodiazepines, is a strong recommendation in the PAD guidelines (Answer D is incorrect). With a QTc of 550 milliseconds, increasing the quetiapine dose for agitation could further increase the QTc and put this patient at high risk

of cardiac arrhythmias. Starting quetiapine would not be the best option (Answer A is incorrect). Starting non-pharmacologic management of agitation/delirium is the best option (Answer B is correct). Dexmedetomidine does not prolong the QTc and should be continued in the setting of agitation (Answer C is incorrect).

8. Answer: D

Dementia is a progressive and chronic state of cognitive impairment that worsens over weeks to months; this differs from the acute onset characteristic of ICU delirium (Answer A is incorrect). Alcohol withdrawal should always be a consideration for patients who become altered or agitated in the ICU. This patient's vital signs are normal, and she is presenting in a hypoactive state; these findings are not typical for acute alcohol withdrawal, and alternative causes for her decline in mental status need to be considered (Answer B is incorrect). Adrenal insufficiency in the ICU usually presents with abnormal laboratory values and/or hemodynamic instability. This patient's laboratory values and hemodynamics are reported as normal; therefore, adrenal insufficiency is unlikely (Answer C is incorrect). Both iatrogenic and non-iatrogenic causes for delirium are well documented in the literature. Dehydration and infection are common causes of delirium, particularly in the older adult population. This patient has several reasons for being dehydrated in the hospital: NPO status, persistent fevers, and taking hydrochlorothiazide. Untreated infection or lack of source control is also a concern with the presence of persistent fever, even while the patient is taking antibiotics (Answer D is correct).